

Chapter 4: The OSINT Collection Methodology

From Requirement to Report to Continuous Watch

Companion Workbook to the Instructional Video

How to Use This Workbook

This is the written companion to the instructional video on the intelligence collection methodology. It works as a follow-along guide, a standalone textbook chapter, and a practice manual.

Where this chapter sits in the series. Chapter 1 covered what intelligence is and introduced the lifecycle in general terms. Chapter 2 covered what OSINT is and why verification is its central skill. Chapter 3 covered the nineteen source categories and how to choose among them. **This chapter puts all of it into motion.** It is the operating manual: how a requirement becomes a collection plan, how collection becomes organized data, how data becomes meaning, how meaning becomes a judgment, how a judgment reaches a decision maker, and how the loop closes and keeps running.

The five labs in this chapter are **deliberately chained**. Lab 1 produces requirements. Lab 2 turns them into a collection plan. Lab 3 executes collection and processing. Lab 4 exploits and analyzes. Lab 5 produces, presents, and revises the report. By the end you will have run one complete investigation from a blank page to a briefed and revised product. Do them in order. Skipping ahead breaks the chain.

Time estimate:

Component	Estimated time
Video	60 to 75 minutes
Reading material	5 to 6 hours
Knowledge assessment	1 hour
Critical thinking exercises	2 hours
Labs 1 through 5 (chained)	12 to 18 hours
Capstone assignment	6 to 9 hours
Total	27 to 37 hours

Materials: a computer with internet access, a word processor, and a spreadsheet application. A partner or small group for the Lab 5 briefing, or an instructor to play the stakeholder role. No paid tools are required.

The Rules Still Apply

Carried forward and enforced in every lab and the capstone.

Rule 1: No private individuals. Fictional entities, organizations, and historical public matters only.

Rule 2: Publicly available means genuinely public. No false identities, no credential use, no defeating access controls, no accepting material from someone unauthorized to share it.

Rule 3: Document as you go. Source, category, URL or citation, access date and time, archived copy, ratings, and coverage boundary for every finding.

Rule 4: No dark web access.

This chapter adds one more, because it is the chapter where you will actually run a collection effort:

Rule 5: Automation is constrained. No scripted or bulk collection against any source whose terms prohibit it, no rate that burdens a small operator's server, and no collection tooling run from an employer's or school's infrastructure without authorization. Section 1.4.6 explains why in detail.

Learning Objectives

Knowledge objectives (you can explain it):

1. Explain why a methodology exists and what specifically fails without one.
2. Name the phases of the OSINT collection methodology in order and state the purpose and output of each.
3. Distinguish an intelligence requirement from a collection requirement, and explain why conflating them produces unusable tasking.
4. Explain the difference between collection, processing, exploitation, and analysis, and why exploitation is treated as its own phase.
5. Describe what a complete collection plan contains and why each element is there.
6. Explain the legal, ethical, and operational security considerations that belong in planning rather than in execution.
7. Describe the processing operations that make collected material usable, and the consequence of skipping each.
8. Explain what dissemination decisions consist of beyond writing the document.
9. Explain why feedback is the most frequently skipped phase and what it costs.
10. Describe how continuous collection differs from episodic collection in structure, not just in duration.

Application objectives (you can do it):

1. Write a priority intelligence requirement that names a decision, is answerable, is deadline-bound, and would change what the consumer does.
2. Decompose a requirement into essential elements of information and then into observable indicators.
3. Build a collection plan that assigns indicators to categories and specific sources, with legal notes, cost, timing, and expected coverage.
4. Execute collection with a documentation standard another person could audit and reproduce.
5. Process a mixed set of collected material into a normalized, deduplicated, entity-resolved, indexed dataset.
6. Exploit processed material to extract findings that were not visible in the raw item.
7. Produce an analytic assessment with confidence ratings, alternatives considered, assumptions, and ranked gaps.
8. Write and brief a professional intelligence product, take feedback, and revise it.
9. Design a continuous monitoring plan with indicators, triggers, thresholds, and a review cadence.

Analysis objectives (you can judge it):

1. Diagnose which phase failed when an intelligence effort produces a bad outcome.
 2. Recognize the common collection errors and predict their downstream effects.
 3. Evaluate whether a collection effort has reached saturation or should continue.
 4. Judge whether a product changed a decision, and if not, why not.
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Part 1: Reading Material

1.1 Introduction to the Intelligence Collection Methodology

1.1.1 Why a Methodology at All

A methodology is a repeatable sequence with defined inputs, outputs, and quality gates. It exists for four reasons, and each corresponds to a specific failure that occurs without it.

Reason	The failure it prevents
Repeatability	Two analysts given the same question produce unrelated work of unrelated quality, and neither can be checked
Auditability	A finding is challenged and nobody can reconstruct how it was reached, so it collapses
Efficiency	Effort is spent on collection that no decision

Reason	The failure it prevents
	depends on, while the decision-relevant gap stays open
Diagnosability	An effort fails and the organization cannot tell which part failed, so it fixes the wrong thing

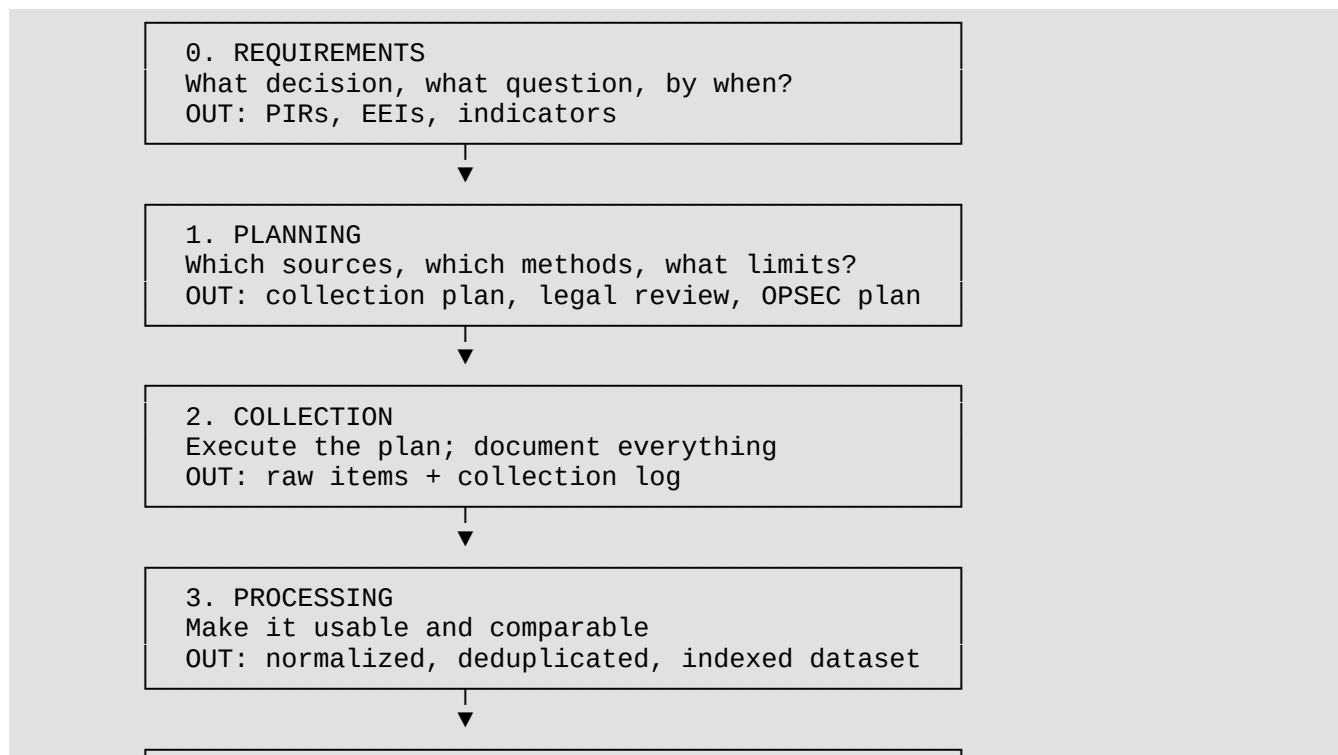
The fourth is the one students underrate. When intelligence goes wrong, the instinct is to collect more. Chapter 2's failure cases showed that in nearly every documented failure the information was substantially available, and the breakdown was in processing, analysis, dissemination, or the consumer relationship. A methodology with named phases lets you locate the break. Without one, every failure looks like a collection failure, and the organization buys more collection.

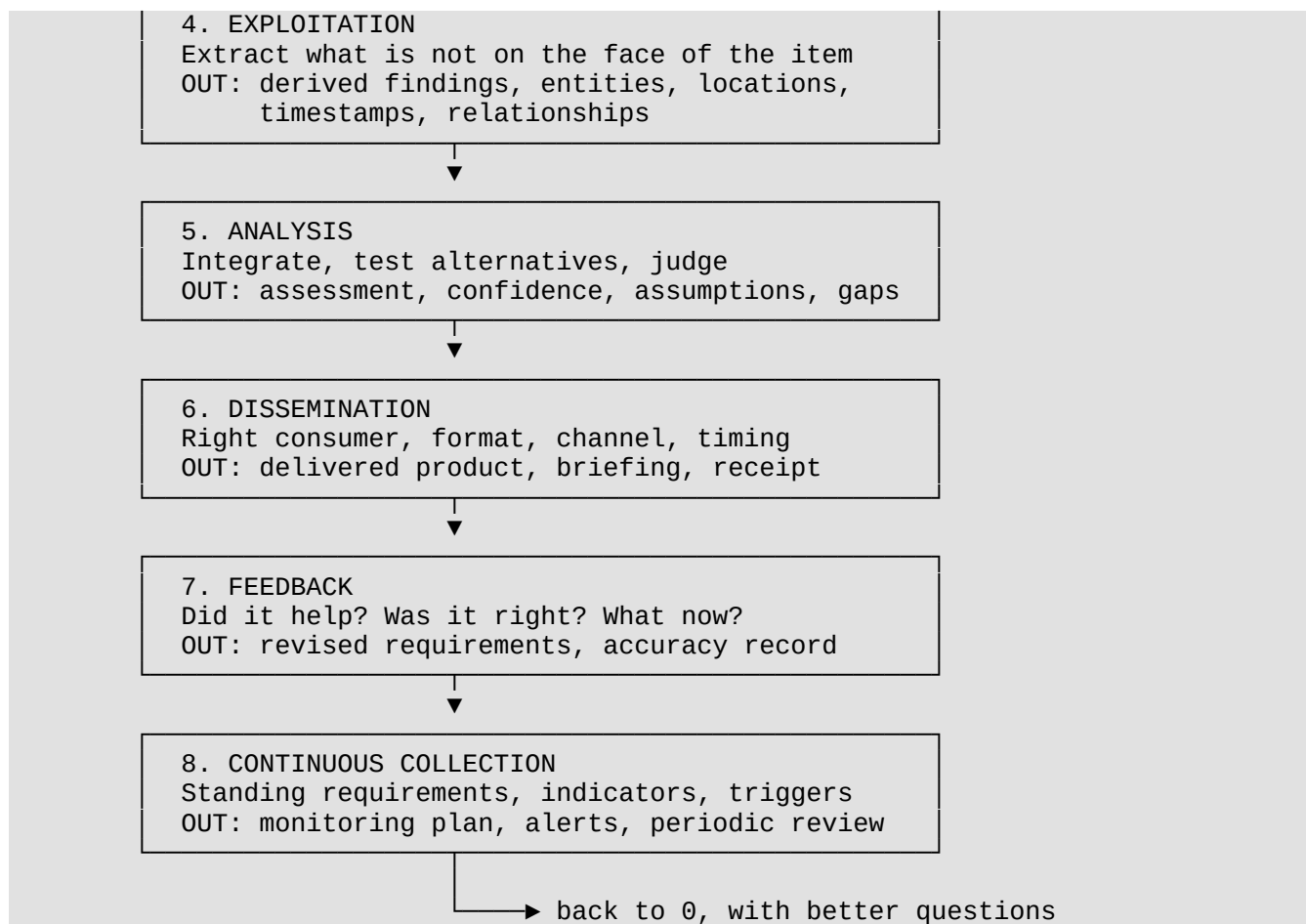
1.1.2 The Methodology, in Full

Chapter 1 taught six phases. This chapter uses nine, because OSINT practice separates two things the general model compresses, and adds one the general model treats as an afterthought.

Exploitation is separated from processing and from analysis. Processing makes material usable. Exploitation extracts findings from a usable item that were not visible on its face: geolocating a photograph, reading metadata, pulling entities out of a document, interpreting an image. Analysis then integrates those findings across items into a judgment. Compressing exploitation into either neighbor causes real problems, described in Section 1.6.

Continuous collection is treated as a phase rather than as "do it again." An investigation ends. A monitoring program does not, and its structure is genuinely different.





1.1.3 The Model Is Linear; Reality Is Not

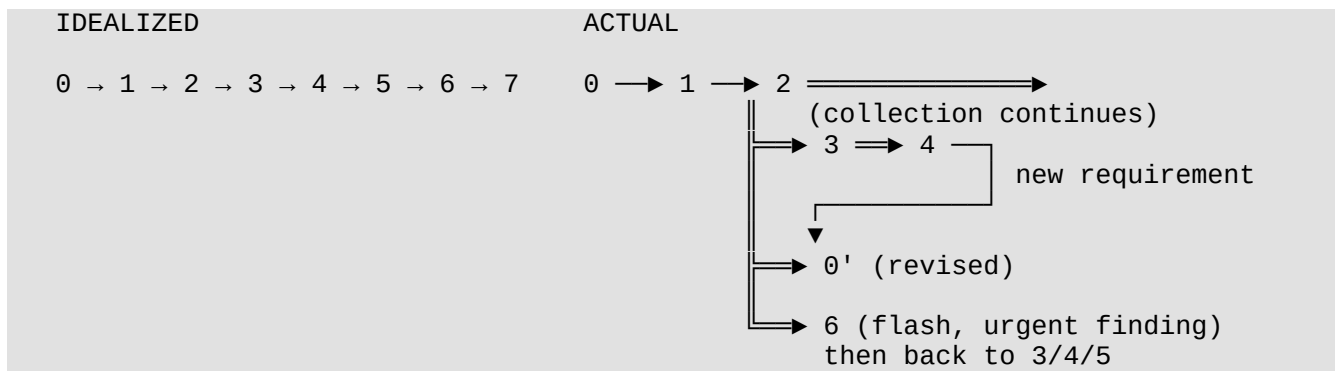
Learn the clean sequence first, because it gives you the vocabulary and the quality gates. Then expect four deviations that are normal rather than pathological:

Phases run in parallel. Collection continues while processing and analysis are underway on what has already arrived.

Findings jump the queue. An urgent discovery in collection goes straight to dissemination as a flash report, then rejoins the sequence.

Phases loop backward. Exploitation of one item frequently generates a new collection requirement, which restarts the sequence for that thread.

The requirement changes mid-effort. The consumer learns something, the decision shifts, and the question you were answering stops being the question that matters. This is not a failure; it is the loop working, and the correct response is to reissue the requirement rather than to keep answering the old one.



1.1.4 The Quality Gate at Each Phase

The methodology's real value is that each phase has a pass condition. Do not proceed until it is met.

Phase	Gate: do not proceed until...
Requirements	The question names a decision, is answerable, has a deadline, and the consumer has endorsed it
Planning	Every indicator is assigned to a category and a specific source, and every line has passed the legal and ethical tests
Collection	Every item is logged with source, timestamp, archive, and ratings, and every negative result carries its coverage boundary
Processing	Timestamps, names, units, and entities are normalized and duplicates are collapsed
Exploitation	Each item's derived findings are recorded separately from the item itself, with the method used
Analysis	At least one alternative explanation has been tested, assumptions are written down, and a confidence level with a stated reason exists
Dissemination	The format matches the consumer, the judgment is in the first sentence, and receipt is confirmed
Feedback	The consumer has answered whether anything changed as a result
Continuous	Indicators, thresholds, owners, and a review cadence are written down

A practical note on discipline. Most students fail these gates in the same two places: they begin collecting before the requirement is answerable, and they begin analyzing before processing is complete. Both feel productive. Both cost more time than they save.

1.1.5 The Methodology Scales Down

A common objection is that this is too heavy for a two-hour question. It is not, because the phases compress without disappearing. Compare a six-week assessment with a ninety-minute one:

Phase	Six-week effort	Ninety-minute effort
Requirements	Negotiated with the consumer over several days	Two minutes: write the question down and confirm it in one message
Planning	Formal plan, legal review, named sources	Five minutes: list four sources and check the four tests
Collection	Dozens of sources, iterative	Forty minutes, logged as you go
Processing	Structured dataset, entity resolution	Ten minutes: one spreadsheet, consistent dates and names
Exploitation	Full imagery and document exploitation	Ten minutes: pull the entities and dates out
Analysis	Formal competing hypotheses, peer review	Fifteen minutes: write the alternative and why you rejected it
Dissemination	Formal report and briefing	Five minutes: four sentences in the channel the consumer reads
Feedback	Structured instrument	One line: "did this answer it?"
Continuous	Monitoring program	One calendar reminder

The ninety-minute version still has nine phases. What compresses is depth, not the sequence. **What you must never do is compress by deleting a phase**, because a deleted phase does not save time; it moves the cost downstream, where it is larger.

1.2 OSINT Data Requirements

1.2.1 The Distinction That Governs Everything

The most consequential vocabulary in this chapter:

An intelligence requirement is what the decision maker needs to know. It is phrased as a question about the world, and it is tied to a decision.

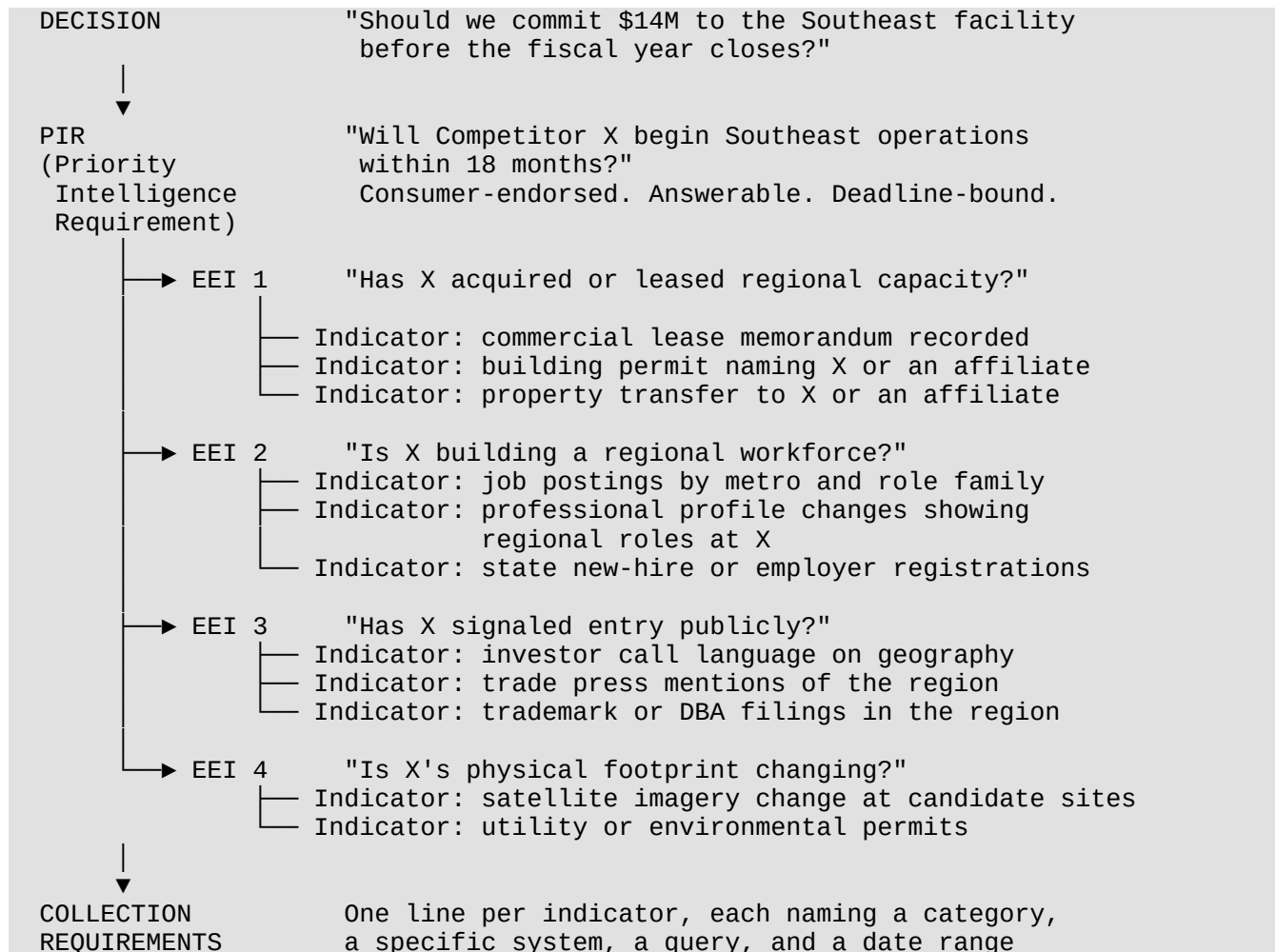
A collection requirement is what a collector must go find. It is phrased as a specific, observable item from a specific source.

They are not the same thing, and the translation between them is the analyst's job. A consumer who hands you a collection requirement has done your job badly, and a collector who receives an intelligence requirement has been given something they cannot execute.

	Intelligence requirement	Collection requirement
Asks	What do I need to know to decide?	What do I need to go get?
Phrased as	A question about the world	A specific retrievable item
Owner	The consumer, endorsed	The collector, tasked
Example	Will this competitor enter our region within 18 months?	Retrieve all commercial lease filings naming the competitor in these four counties for the past 12 months
Test of a good one	Answering it changes a decision	It can be executed by someone who knows nothing about the case
Failure if wrong	Effort answers a question nobody had	Collector cannot act, or collects the wrong thing

1.2.2 The Decomposition Chain

The full chain runs from a decision down to a thing you can go look at.



Each level answers a different question. The decision says why anyone cares. The PIR says what would change the decision. The EEIs say what must be true for the PIR to resolve. The indicators say what you could actually see. The collection requirements say where to go look.

The most common student error is to jump from the PIR straight to searching. That produces browsing, not collection, and it cannot be audited because there was never a plan to audit it against.

1.2.3 Writing a Requirement That Works

Four tests, from Chapter 1, restated with the failure each catches:

Test	Question	Failure it catches
Decision-linked	What decision does this support, and who makes it?	"Tell me about X" requirements that answer nothing
Answerable	Could any answer be reached from available sources?	Requirements about private intent that no open source reaches
Deadline-bound	When is the decision made?	Perfect analysis delivered after the vote
Consequential	Would a yes and a no lead to different actions?	Questions whose answer changes nothing

Apply all four. A requirement that passes three and fails one is not a requirement yet.

Worked rewrites:

As received	Problem	Rewritten
"Everything on Company X"	No decision, no bound, not answerable	"Does the public record contradict Company X's representations regarding litigation, ownership, and facility count, ahead of the 14 March board vote?"
"Are we going to be attacked?"	Not answerable; asks about intent	"Which threat actor categories have publicly documented targeting of organizations in our sector and size band in the past 24 months, and which of their documented initial access methods are viable against our currently exposed assets? Needed before the 1 June budget submission."
"Is the neighborhood safe?"	Undefined term, no decision	"What do the past 36 months of published incident and code

As received	Problem	Rewritten
		enforcement records show for the four-block area around 1170 Quarry Line Road, and how does that compare with the county rate? Needed before the 9 April site selection meeting."
"Find out what the competitor is doing"	No bound, no decision	Decompose into the chain in Section 1.2.2

1.2.4 Prioritization Is Part of the Requirement

Resources are always finite, so an unranked set of requirements is not a set of requirements; it is a wish list. Rank explicitly using two dimensions:

	High decision impact	Low decision impact
Answerable from open sources	Do first. This is where the value is	Do only if cheap, or defer
Not answerable from open sources	Escalate: name the non-open source or process that would answer it, and tell the consumer early	Drop, and say you dropped it

The top-right and bottom-left quadrants are where professionalism shows. Telling a consumer in week one that their most important question cannot be answered from open sources is far more valuable than discovering it in week five.

1.2.5 Requirement Documentation

Every requirement gets a record with these fields, because the record is what makes the whole effort auditable later:

Field	Content
Requirement ID	A short reference used in every downstream artifact
Decision supported	The decision, the decision maker, and the date
PIR statement	One sentence
EEIs	Numbered, each traceable to the PIR
Indicators	Numbered, each traceable to an EEI
Priority	Rank, with the reason
Deadline	Tied to the decision

Field	Content
Answerability assessment	Which parts open sources can and cannot reach
Endorsement	Who approved it and when
Revision history	Because requirements change, and the change must be visible

That last row matters more than it looks. When a requirement changes mid-effort, undocumented drift means the final product answers a question nobody endorsed.

1.2.6 What Requirements Cannot Do

Two honest limits, both worth saying to a consumer early.

Requirements cannot make an unanswerable question answerable. Open sources rarely reach private intent, internal deliberation, or unexpressed plans. Chapter 2 established this and it does not change here. The professional move is to reframe: if you cannot answer "does the competitor intend to enter," you can often answer "what observable steps would necessarily precede entry, and have any occurred."

Requirements cannot substitute for a decision that has not been made. If the consumer does not actually know what they would do differently, no requirement will help, and the honest response is to work that out with them before collecting anything.

1.3 Planning OSINT Collection

1.3.1 What a Collection Plan Is

A collection plan is the document that converts indicators into executable tasking. It is not a list of websites. It is a table in which every row can be handed to a person who knows nothing about the case, and that person can execute it, know when they are done, and know what to record.

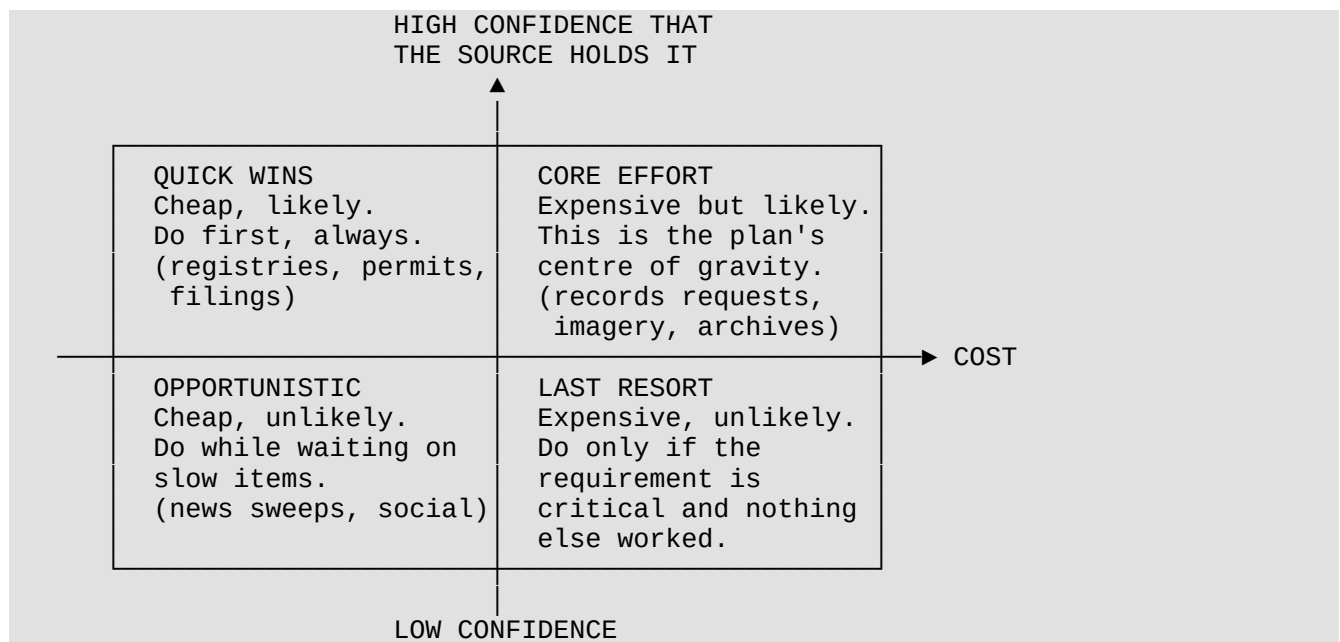
The anatomy of a plan row:

Element	Why it is there
Indicator ID	Traceability back to an EEI and a PIR. Every row exists because a requirement demanded it
Category	From Chapter 3. Forces category-first thinking rather than tool-first
Specific source or system	"County recorder grantor/grantee index" not "public records"
Query or approach	The exact terms, name variants, and filters to use
Date or scope range	Bounds the work and defines completeness

Element	Why it is there
Access model and cost	Free, fee, registration, in person, request
Legal and ethical note	The four tests, applied per row, before anything is collected
Expected coverage boundary	What this source will and will not cover, written in advance
Expected latency	How long until the material is available
Owner	Who executes it
Priority	High, medium, low, keyed to requirement priority
Status	Not started, in progress, complete, blocked
Output location	Where the collected material and log entry will live

1.3.2 The Collection Emphasis Matrix

Before writing rows, decide where effort goes. A simple matrix keeps a plan from becoming an undifferentiated list:



The practical rule: **start with quick wins to establish the frame, then spend the bulk of the effort in the core quadrant, and use opportunistic collection to fill the waiting time while slow items (records requests, imagery orders, interlibrary loans) are in flight.** Slow items should be initiated on day one precisely because they are slow.

1.3.3 The Legal and Ethical Review Belongs Here

The single most important structural point in this section: **the legal and ethical review happens in**

planning, not in execution. Reviewing after collection is not a review; it is a post-mortem.

Every row of the plan passes the four tests from Chapter 3:

Test	Question	If it fails
Access	Could any member of the public obtain this lawfully?	Remove the row
Identity	Can I obtain it without misrepresenting who I am?	Remove the row
Protection	Am I defeating any control the owner placed?	Remove the row
Purpose and harm	Legitimate purpose, proportionate foreseeable harm?	Remove or narrow the row, and record the decision

Plus three planning-specific additions:

The prohibited-activity statement. Written before collection begins, listing what this engagement will not do. Deciding under deadline pressure is how people end up somewhere they would not have chosen calmly.

The individuals policy. Which individuals may appear in this effort, in what capacity, and what will not be collected about them. For most engagements the correct policy is: individuals appear only in a professional or official capacity relevant to a stated requirement, nothing further, no family members, no minors.

The escalation path. Who decides when a row is unclear. Every plan should name a person, and "I will ask before I collect" should be the default for anything ambiguous.

1.3.4 Operational Security for the Collector

Planning must account for the fact that collection is sometimes visible to the subject.

Consideration	Practical measure
Site visit logs	Assume the subject can see visits to their own site. Space out, use archives where possible
Platform view notifications	Some platforms notify subjects of profile views. Know which, and decide in advance
Records request visibility	In many jurisdictions a request is itself a public record. Assume the subject may learn of your interest
Attribution to your organization	Do not research from a network that identifies your employer if that matters to the engagement
Account use	Use a research browser profile not signed in to personal accounts. Never a false identity

Consideration	Practical measure
Rate and volume	Automated or rapid access is detectable and may breach terms
Data handling	Decide before collecting where material is stored, who can see it, and when it is purged

The boundary that matters: OPSEC protects the integrity of a lawful inquiry. It is never a justification for doing something you would not do openly. If concealment is what makes a collection possible, the collection is the problem.

1.3.5 Resourcing and Sequencing

Three planning judgments that beginners skip:

Initiate slow items first. Records requests, imagery orders, interlibrary loans, and fee-based document retrieval have latency measured in days or weeks. If the deadline is three weeks, these go out on day one or they will not arrive.

Budget the time honestly. A useful rough split for a first-time effort: 15% requirements and planning, 30% collection, 20% processing, 15% exploitation, 15% analysis, 5% dissemination. Students consistently under-budget processing and then discover on the last day that their material cannot be compared.

Define the stopping condition in advance. Write down, before collecting, what "done" looks like for each requirement: a specific set of systems searched, a specific date range covered, and a saturation test (see Section 1.4.7). Without a stopping condition, collection expands to fill all available time.

1.3.6 A Worked Collection Plan Extract

For EEI 1 from Section 1.2.2 ("Has X acquired or leased regional capacity?"):

Ind.	Category	Specific source	Query / approach	Range	Access	Legal note	Coverage boundary	Latency	Priority
1.1	Government records	County recorder, grantor/grantee index, 4 counties	Search entity name plus 6 variants and 2 known affiliates	24 months	Free view, fee per copy	Public index; passes all four tests	Only these 4 counties; other counties unchecked	Same day	High
1.2	Government records	County building permit	Applicant name variants;	24 months	Free	Public; passes	Permits only, not leases	Same day	High

Ind.	Category	Specific source	Query / approach	Range	Access	Legal note	Coverage boundary	Latency	Priority
		portals, same 4 counties	also search by parcel for candidate sites						
1.3	Corporate filings	State registry, foreign qualification records	Entity plus variants; check for new qualification in the region	24 months	Free	Public; passes	Registrations only; does not prove operations	Same day	High
1.4	Newspapers	Legal notices in 4 county papers of record	Fictitious business name and lease-related notices	24 months	Library database	Public; passes	Only papers of record; smaller outlets missed	1 day	Medium
1.5	Satellite imagery	Commercial imagery over 3 candidate sites	Current plus archive at 6-month intervals	24 months	Fee	Public commercial purchase; passes	Cloud cover may limit dates	2 to 5 days	Medium
1.6	Archived websites	Captures of X's locations page	Enumerate captures; compare location lists	36 months	Free	Public; passes	Capture intervals limit dating precision	Same day	High

Notice what this table makes possible: someone else can execute it, the coverage boundaries are written before the searching rather than after, the fee and latency items are visible so they can be started early, and every row traces back to a requirement.

1.4 OSINT Collection

1.4.1 Execution Discipline

Collection is the phase students find easiest and do worst, because the work feels like searching and the discipline feels like overhead. Four rules make the difference.

Work the plan, and record deviations. If you depart from the plan, note it and why. Drift is normal; undocumented drift is untraceable.

Log at the moment of collection. Not at the end of the day, and never at the end of the project. Retroactive logging produces missing timestamps and invented certainty.

Archive at the moment of collection. Web content changes and disappears. A finding you cannot re-retrieve is a finding you cannot defend.

Collect the negative results too. A search that returns nothing is a data point with real value, and it must be recorded with its coverage boundary or it is worthless.

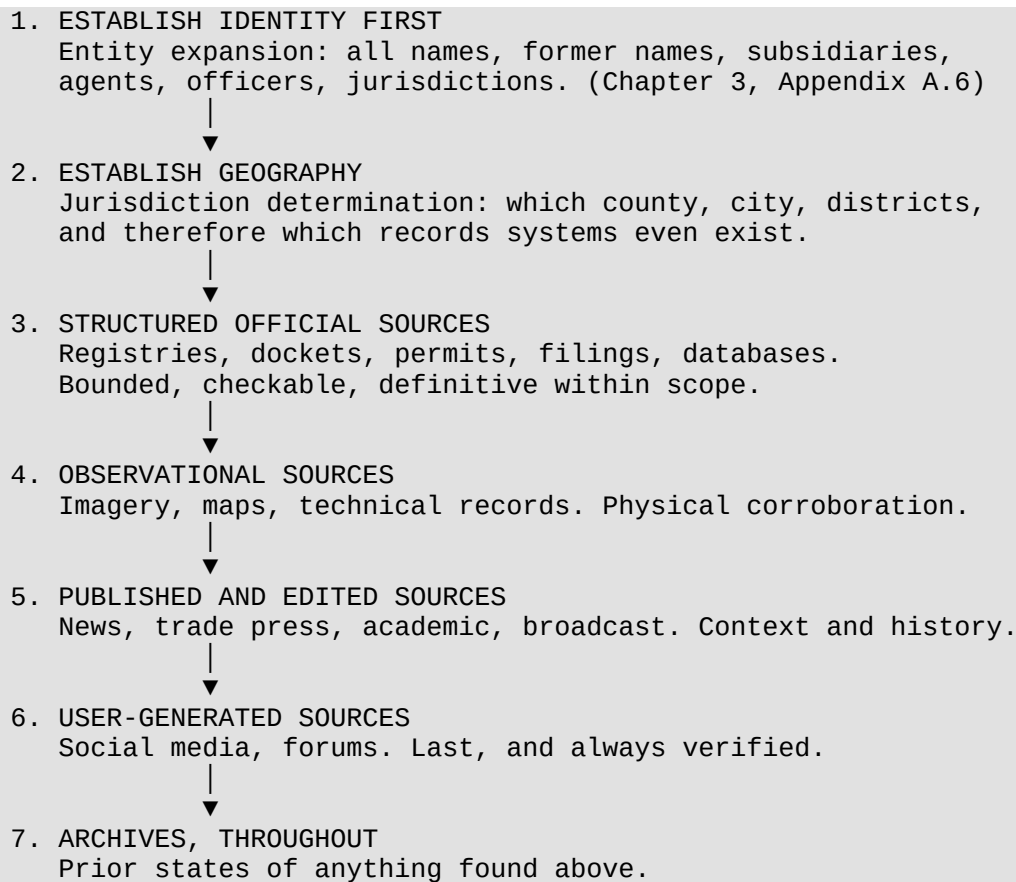
1.4.2 The Collection Log Standard

Carried forward from Chapter 2 and extended for this chapter:

Field	Why
Item ID	Reference used in processing, exploitation, analysis, and the product
Requirement ID / Indicator ID	Traceability: why was this collected?
Claim or content summary	One sentence, so the log is scannable
Category	From Chapter 3
Source or system name	Specific
URL or citation	Reproducibility
Query used	So another person can repeat the search exactly
Access date and time, with time zone	Because content changes
Archived copy location	Because links rot
Publisher type and apparent independence	Circular reporting detection
Reliability (A to F)	Chapter 1 framework
Credibility (1 to 6)	Chapter 1 framework
Coverage boundary	Especially for negative results
Collector initials	Accountability

1.4.3 Collection Order Matters

There is a right sequence within a collection effort, and it is not obvious.



Why this order. Steps 1 and 2 determine what you are even able to search; getting them wrong invalidates everything downstream. Structured official sources come next because they are bounded and definitive, and they give you the frame against which everything else is judged. User-generated material comes last because it is the most likely to mislead and because, by the time you reach it, you have a frame that lets you evaluate it.

Students who run this in reverse (social media first, records last) produce investigations that are confidently wrong, because the early unverified material shapes how everything later is read. That is anchoring, and the sequence above is a structural defense against it.

1.4.4 Manual and Automated Collection

	Manual	Automated
Method	A person queries, reads, judges, records	Scripts, feeds, alerts, or scheduled retrieval
Best for	Judgment-heavy work: reading a docket, interpreting imagery, evaluating a claim	Repetitive, high-volume, or continuous work: monitoring, bulk downloads, alerting
Throughput	Low	High
Judgment applied	At every item	Only where encoded in advance
Error mode	Fatigue, inconsistency, missed	Systematic error repeated at

	Manual	Automated
	items	scale; silent breakage
Documentation	Manual discipline required	Logged automatically, which is an advantage
Legal exposure	Low	Higher: terms of service, rate limits, computer misuse questions
Reproducibility	Depends on logging	High, if the script and parameters are retained
Right use in this course	Everything in the labs	Only where permitted, disclosed, and rate-limited

Rule 5 in practice. Automation is a legitimate professional technique and it is constrained in this course for good reasons: terms of service frequently prohibit it, the legal position on scraping continues to develop, aggressive automated access can burden small operators, and a script running against the wrong system from an institutional network is an incident. Where automation is used professionally, the discipline is: check the terms, respect published access policies, rate-limit deliberately, identify your client honestly, retain the script and its parameters as part of the collection record, and have written authorization.

The silent-breakage problem deserves emphasis. An automated collector that stops returning results usually fails quietly. Weeks later an analyst concludes that activity stopped, when in fact collection stopped. Every automated collection needs a heartbeat check: an alert when *nothing* arrives, not only when something does.

1.4.5 Common Collection Errors

Error	What it looks like	Downstream impact
Searching before the requirement is answerable	Browsing with a deadline	Everything collected is unusable because nothing was asked
Wrong entity	Searching the name you were given only	Material findings under former names and subsidiaries never surface
Wrong jurisdiction	Searching the state when the record is at the county	False negative treated as a clean result
No coverage boundary	"I could not find anything"	A limitation is reported as a finding
Retroactive logging	Building the log at the end	Timestamps invented, queries misremembered, work unauditabile
No archiving	Citing a live URL	Finding evaporates when the page changes

Error	What it looks like	Downstream impact
Single-category corroboration	Three articles agreeing	Circular reporting mistaken for confirmation
Collection drift	Following an interesting thread with no requirement	Time spent, requirement still open
Over-collection	Thousands of items, no triage	Processing and analysis capacity swamped
Anchoring on early material	First item shapes all reading	Confirmation bias baked in before analysis begins
Terms of service breach	Bulk access to a prohibited source	Findings unusable, organization exposed
Individual drift	Requirement about a company, collection about a person	Ethical breach and legal exposure

1.4.6 Stopping: The Saturation Test

Collection does not end when you run out of curiosity. It ends when one of four conditions is met, and you should write down which one applies:

Requirement satisfied. Every indicator has been checked in its planned source, and the EEIs resolve.

Saturation reached. New collection is returning material you already have. A practical test: if the last ten items produced no new entity, no new date, and no new relationship, you are saturated for that thread.

Deadline reached. Time is up. This is a legitimate stopping condition and it must be reported as one, with the gaps it leaves.

Blocked. The remaining indicators cannot be collected from open sources. Report this early, not at the end.

What is not a stopping condition: feeling like you have enough, or finding something that confirms your initial theory. Both are how efforts end prematurely with a wrong answer.

1.4.7 Collection Ethics in Execution

Three situations arise during collection that planning cannot fully anticipate, and each has a correct handling:

You encounter material that was not meant to be public. An exposed document on a misconfigured server, a file left in an open directory. Do not collect it. Note that you encountered it, do not distribute it, and where appropriate notify the owner. Availability is not authorization.

You encounter information about an individual outside the requirement. Do not follow it. Record in the ethics annex that you encountered it and did not pursue it. This is a positive act of professional discipline and it should be visible in your documentation.

Someone offers you material. A tip, a document, an insider account. Verify what it is and where it came from before touching it. Material from a person not authorized to share it is outside OSINT, and the analytic quality of the material is irrelevant to that determination.

1.5 OSINT Data Processing

1.5.1 What Processing Is For

Processing converts a pile of collected items into a dataset that can be compared. It adds no meaning. It performs no interpretation. Its entire job is to make the material *comparable*, and it is the phase students skip most often and regret most reliably.

The test of complete processing: **could a second analyst, given only your processed dataset, sort it by date, group it by entity, and identify duplicates without asking you anything?** If not, processing is not done.

1.5.2 The Processing Operations

Operation	What it does	Consequence of skipping
Normalization: time	All timestamps to one standard with time zones recorded	False event sequences; causal chains that never happened
Normalization: names	Consistent format for people, entities, places	Entities appear as several; relationships invisible
Normalization: units and currency	Consistent measures	Meaningless comparisons
Normalization: identifiers	Consistent format for case numbers, parcel IDs, registration numbers	Records that should join do not
Deduplication	Recognizing that several items describe the same thing	Inflated corroboration; one wire story counted five times
Entity resolution	Determining that variant references are one entity, or that identical references are two	Under-counted activity, or invented conspiracies
Translation	Foreign-language material rendered, with the original retained	Whole source populations excluded
Transcription	Audio and video rendered as searchable text	Hours of unusable material
OCR	Scanned documents made searchable	Archives and older records effectively unreadable
Structuring	Extracting fielded data from unstructured items into a table	Nothing can be sorted, counted, or compared

Operation	What it does	Consequence of skipping
Tagging	Applying a controlled vocabulary: requirement ID, entity, location, date, type, reliability	Retrieval becomes impossible at volume
Indexing and storage	A retrievable, backed-up location with consistent naming	Material collected and then lost

1.5.3 A Minimum Viable Processing Schema

You do not need a database. A single spreadsheet with these columns handles most student and small-team investigations:

Column	Content
Item ID	Sequential, referenced everywhere downstream
Requirement / Indicator ID	Why this exists
Item type	Filing, docket entry, article, image, post, record, dataset
Category	Chapter 3 category
Source system	Specific
Event date (normalized)	The date the <i>thing described</i> occurred, in ISO format
Publication date (normalized)	The date the item was published
Access date and time	With time zone
Entities (resolved)	Canonical names, semicolon-separated
Location (normalized)	Address, coordinates, or jurisdiction
Duplicate of	Item ID, where applicable
Independence	Original, derivative, or unknown
Reliability / Credibility	A-F and 1-6
Archive location	Path or URL
Coverage boundary	For negative results
Processing notes	What was changed and why

Two dates, not one. The distinction between event date and publication date is the one students collapse, and collapsing it destroys timelines. An article published in March describing a lease signed in January is a January event and a March publication, and a timeline built on publication dates would put the lease in the wrong quarter.

1.5.4 Entity Resolution in Practice

The highest-value processing operation, and the one with the most judgment in it.

The problem in both directions. Splitting one entity into several hides a pattern. Merging several into one invents a relationship. Both are serious errors and they require opposite corrections.

A working procedure:

1. **Build a canonical list.** One row per real-world entity, with a chosen canonical name.
2. **Record every observed variant** against its canonical entity: legal name, trade names, former names, abbreviations, misspellings, transliterations.
3. **Require a second identifier before merging.** A name match alone is never enough (Chapter 3). Acceptable second identifiers: registration number, matching address, shared officer, case number linking them, or an explicit statement in a filing.
4. **Record the basis for every merge.** "Merged on shared registered agent address and matching manager name" is auditable. A silent merge is not.
5. **Flag unresolved candidates rather than guessing.** An "unresolved: possible same entity" flag is a legitimate output and becomes a collection requirement.

1.5.5 The Master Timeline

The single most useful processed artifact in most investigations. One row per dated event, sorted, with sources attached.

Field	Content
Date (normalized)	ISO format; use a range where the date is bracketed
Precision	Exact, day, month, quarter, or bracketed window
Event	One sentence, stated as observed
Entities involved	Canonical names
Source item IDs	Every item supporting this row
Categories supporting	How many distinct categories, from Chapter 3
Confidence in the event	Whether the event itself is established
Notes	Conflicts, gaps, questions raised

The precision column is the professional touch. A timeline that renders a bracketed change ("between March and July") as a specific date is lying quietly. Recording precision alongside the date prevents that, and it makes gaps visible: a timeline with three exact dates and eleven month-level entries tells you where more collection would help.

1.5.6 Processing Quality Gates

Before moving to exploitation:

- Every item has a normalized event date and publication date, or an explicit note that no date could be established.
- Every entity reference maps to a canonical entity, or is flagged unresolved.
- Duplicates are marked, not deleted, with the original retained.
- Every item is tagged to at least one indicator.
- Every archived copy is retrievable from the recorded location.
- The master timeline exists.

Passing these gates typically takes longer than students expect and saves more time than they believe.

1.6 OSINT Data Exploitation

1.6.1 Why Exploitation Is Its Own Phase

Exploitation extracts from a processed item information that is not on its face. It sits between processing (which makes items comparable) and analysis (which integrates across items to reach a judgment).

The three are genuinely different operations on different objects:

Phase	Operates on	Produces	Question answered
Processing	The item's form	A comparable, indexed item	"Can I sort and join this?"
Exploitation	A single item's content	Derived findings from that item	"What does this item contain that I could not see?"
Analysis	The set of all derived findings	A judgment	"What does all of it mean, and how sure am I?"

Why the separation matters practically. Two failures follow from collapsing exploitation into its neighbors.

Collapsed into processing, exploitation gets treated as clerical and gets rushed. Geolocation, imagery interpretation, and metadata reading are judgment-heavy and error-prone, and rushing them produces confident, wrong derived facts that then propagate through analysis as though they were observations.

Collapsed into analysis, exploitation's derived findings get mixed with integrative judgments in a single undifferentiated mush. When a reader later asks "how do you know the photo was taken at that

location," the answer must be a documented exploitation step with a stated method, not a general appeal to the analysis.

The recording rule: a derived finding is recorded separately from its source item, with the method used and the confidence in the derivation. "Item 44 (photograph) geolocated to 1170 Quarry Line Road by matching building profile, fence line, and utility pole spacing against street-level imagery; high confidence" is exploitation done correctly.

1.6.2 The Exploitation Operations

Operation	Applied to	Extracts	Method notes
Geolocation	Photographs, video	Where it was captured	Match fixed features against street-level and satellite imagery; record which features
Chronolocation	Photographs, video	When it was captured	Shadow direction and length, weather records, vegetation state, visible dated objects
Metadata extraction	Any file	Device, timestamps, authorship, software, coordinates, relationships	Record the extraction method; corroborating, never conclusive (Chapter 3)
Imagery interpretation	Satellite, aerial, ground imagery	Structures, activity, change, dimensions	Use a known reference for scale; separate observation from interpretation
Document exploitation	Filings, reports, contracts	Named entities, dates, amounts, obligations, cross-references to other documents	The cross-references are frequently the most valuable output
Entity extraction	Text of any kind	People, organizations, addresses, identifiers	Feed straight into the canonical entity list
Relationship extraction	Any item	Who is connected to whom, and how	Record the stated basis for each edge
Financial extraction	Filings, budgets, registers	Amounts, counterparties, obligations, timing	Watch definitions and fiscal boundaries
Technical enrichment	Domains, certificates, addresses	Registration, resolution history, related infrastructure	Bounded and reproducible; a strong category
Transcript exploitation	Audio, video, captions	Statements, claims, timestamps, speaker attribution	Always listen to the passage in context before quoting

Operation	Applied to	Extracts	Method notes
Cross-reference resolution	Any document citing another	The cited document itself	The highest-yield operation in records work, and the most skipped

1.6.3 The Cross-Reference Rule

Worth its own subsection because it is where most of the value in document-heavy investigations sits.

Official documents cite other documents. A permit references an application number. A docket entry references an exhibit. A filing references a prior filing. An inspection report references a prior violation. A consent order references a compliance schedule.

Every cross-reference is a collection requirement. The correct exploitation behavior is to extract each one, add it to the collection plan as a new row, and go get it. Students who read documents in isolation extract perhaps a third of the available value; students who chase cross-references consistently produce investigations that look, to an observer, like they had inside access.

1.6.4 Observation Versus Interpretation

The discipline that keeps exploitation honest. Every exploitation output is written in two parts:

	Observation	Interpretation
Imagery	"A rectangular structure approximately 60 by 30 metres appears in the August image and is absent in the March image"	"A warehouse was constructed between March and August"
Metadata	"The creating application field reads a consumer spreadsheet product; the document contains no template reference"	"The document's claimed origin inside an enterprise environment is not supported"
Document	"The consent order's compliance status field reads 'administratively resolved upon entity dissolution'"	"The compliance obligation was never demonstrated as completed"
Geolocation	"Building profile, fence line, and pole spacing match the frontage at the subject address"	"The photograph was taken at the subject facility"

Write the left column as fact and the right column as assessment with a confidence level. This separation is what allows a later reviewer to accept your observation while disputing your interpretation, which is exactly the kind of disagreement a professional product should be able to survive.

1.6.5 Exploitation Quality Gates

Before moving to analysis:

- Every derived finding is recorded against its source item ID.
 - Every derived finding names the method used to derive it.
 - Every derived finding carries its own confidence, separate from the source item's ratings.
 - Observation is separated from interpretation in every case.
 - Every cross-reference encountered has been added to the collection plan or explicitly closed with a reason.
-

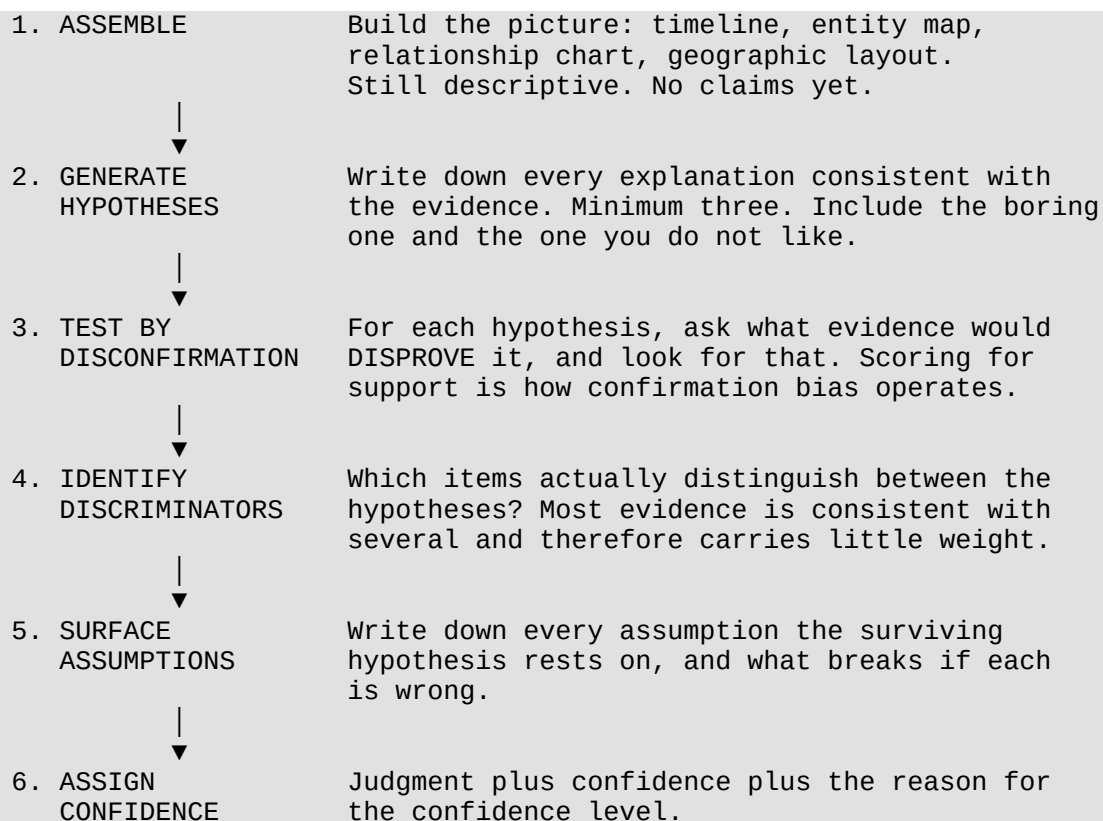
1.7 OSINT Data Analysis

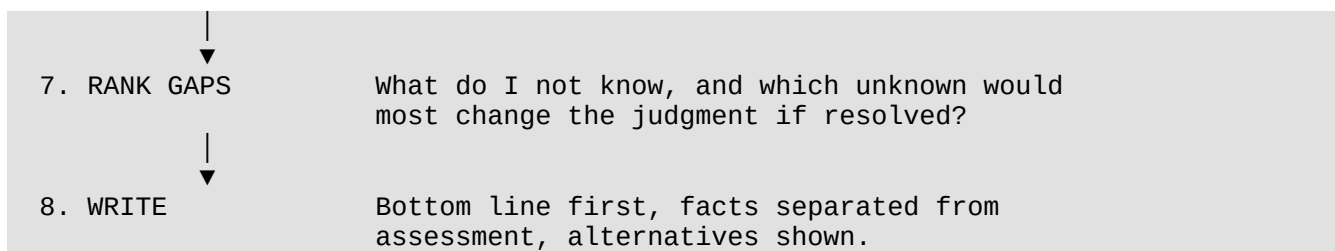
1.7.1 What Changes Here

Analysis is the first phase that produces a claim about the world that no single item states. Everything before it was preparation.

Its inputs are the processed dataset, the derived findings from exploitation, and the master timeline. Its outputs are a judgment, a confidence level with a stated reason, the assumptions the judgment rests on, the alternatives considered and rejected, and the ranked gaps.

1.7.2 The Analytic Sequence





Step 4 is the one that separates trained analysts from careful readers. In most investigations, the large majority of collected evidence is consistent with every hypothesis and therefore has no discriminating power. Identifying the handful of items that actually distinguish between explanations tells you where your judgment really comes from, and it usually reveals that a confident conclusion rests on two or three items rather than on the forty you collected.

1.7.3 Structured Techniques, Applied

Carried from Chapter 1, with the OSINT-specific note for each:

Technique	Use it when	OSINT-specific note
Analysis of Competing Hypotheses	One explanation feels obvious	Score for disconfirmation; open source volume makes confirmation trivially easy to manufacture
Key Assumptions Check	Before any high-stakes judgment	Coverage assumptions are the ones OSINT analysts miss: "the record would exist if it happened" is an assumption
Devil's advocacy	The team converged fast	Assign it explicitly, and require the advocate to use the same dataset
Indicators and warnings	Ongoing monitoring	Define indicators before the event, or you will recognize them only afterward
Red teaming	Forecasting an actor's behavior	Model their incentives, not yours
Link and network analysis	Structure and relationships matter	Record the basis for every edge; unsupported edges propagate
Timeline and pattern analysis	Sequence, cause, and anomaly	Precision must be carried through, not smoothed away
Deception detection	Material arrives conveniently	Ask who benefits from your believing it, and how cheaply it could have been staged

1.7.4 Confidence, Restated for This Phase

Confidence describes how solid the basis for a judgment is. It is not the probability of the event. Both

appear in a good sentence, and they are different measures.

Confidence	Basis in an OSINT context
High	Multiple independent categories agree; sources with strong production processes; few assumptions; derived findings independently checkable
Moderate	Credible sourcing with real gaps; some reliance on a single category; plausible alternatives remain open
Low	Single-category or single-source; heavy assumption load; key derived findings unverified; significant alternatives untested

The category-count heuristic. In open source work, a useful and defensible way to explain a confidence level is to state how many *distinct categories* support the finding and whether they could have been coordinated. "High confidence: supported by a compelled filing, a physical image, and a court exhibit, which have unrelated production processes" explains itself. "High confidence: seven sources" does not.

1.7.5 Bias Controls Specific to OSINT

Bias	How it shows up in OSINT	Control
Confirmation	Searching until you find agreement, then stopping	Predefine the stopping condition and the disconfirming evidence you would look for
Availability	Concluding from what is easy to find	State coverage gaps explicitly; remember that repressive and disconnected environments under-publish
Anchoring	The first item read shapes everything after	Collection order (Section 1.4.3): official records before user-generated content
Circular corroboration	Counting derivative reports as independent	Trace to origin; count categories, not items
Layering	Yesterday's low-confidence judgment becomes today's background fact	Carry confidence forward with the claim; cite your own prior product
Narrative pull	The tidy story fits, so the untidy evidence is set aside	Require the untidy items to be listed and addressed explicitly
Volume equals truth	A lot of material feels like a lot of evidence	Report discriminating items separately from supporting items

1.7.6 Analysis Quality Gates

Before moving to dissemination:

- At least two alternative explanations were written and tested, not just named.
 - The discriminating items are identified.
 - Assumptions are written down with consequences.
 - A confidence level exists with a stated reason.
 - Gaps are ranked by analytic impact and each names the source that would resolve it.
 - No assessment is stated as a fact anywhere in the draft.
-

1.8 OSINT Data Dissemination

1.8.1 Dissemination Is a Set of Decisions, Not a Document

Writing the product is one part. The decisions around it are the rest, and they are what determine whether the work has any effect.

Decision	Options	Failure if wrong
Consumer	Who is this for, precisely	Written for nobody in particular, read by nobody
Format	Flash, one-page brief, full assessment, table, briefing, dashboard entry	A twenty-page assessment for a consumer with six minutes
Length	Sentences to pages	Judgment buried on page nine
Channel	The channel they actually read	Correct product, unread inbox
Timing	Before the decision, with room to act	History rather than intelligence
Handling	Who may receive it, who may not, how it may be shared	A finding that tips a subject or creates legal exposure
Retention	How long it is kept, and what is purged	Records retained past their purpose
Receipt	Confirmed or assumed	Nobody knows whether it landed

1.8.2 Matching Product to Consumer

Consumer	Format	Length	What they need in the first sentence
Executive or board	One-page brief	1 page	The judgment, the confidence, and the

Consumer	Format	Length	What they need in the first sentence
			decision it bears on
Operational lead	Assessment	2 to 5 pages	The judgment plus what changes in their plan
Operator or field staff	Flash or bulletin	1 paragraph	What to do right now
Legal or compliance	Assessment with full sourcing	Variable, plus annexes	The characterization, precisely worded, with the record cited
Technical team	Assessment plus structured indicator list	Variable	What to detect, block, or check
Public body or public record	Formal report	2 to 5 pages, plus annexes	The judgment, phrased knowing both sides will quote it

1.8.3 The Standard Product Structure

Carried forward and now standard for the rest of this chapter:

BOTTOM LINE	Judgment, confidence, estimative language, so-what. 3 to 4 sentences maximum.
KEY FACTS	Observed only. Each with category, source, reliability and credibility ratings.
ASSESSMENT	What it means. Alternatives considered and why rejected. Discriminating evidence named.
KEY ASSUMPTIONS	Each with what changes if it is wrong.
INTELLIGENCE GAPS	Ranked by analytic impact; each names the source that would resolve it.
RECOMMENDED ACTIONS	Specific, assignable, time-bound.
WHAT WOULD CHANGE THIS ASSESSMENT	2 to 4 sentences.
COLLECTION LOG	Annex; does not count toward length.
ETHICS ANNEX	Annex; individuals encountered and handling.

1.8.4 Briefing the Product

Dissemination frequently includes speaking to it, and Lab 5 requires it. The structure that works:

Open with the judgment. Not the background, not the methodology, not what you searched. The judgment and the confidence, in one sentence, within the first fifteen seconds.

Give the two or three discriminating facts. The ones that actually drove the judgment, not the full evidence base.

State the alternative you rejected and why. This is what earns credibility with a skeptical audience, because it demonstrates that you looked for reasons to be wrong.

Name the gaps and what would close them. This converts the briefing from a verdict into a decision aid.

Stop talking. The remainder of the time belongs to the consumer's questions, and the questions are where you learn what the requirement actually was.

Answering questions well. Three habits: say "I do not know" when you do not, distinguish out loud between what you observed and what you assessed, and when challenged on a finding, cite the item and the method rather than defending the conclusion. A briefer who cannot say "that is an assessment, and here is the assumption it rests on" will eventually be caught out.

1.8.5 Writing for a Product That Will Be Quoted

Increasingly, OSINT products are read by parties with an interest in the outcome. Four disciplines:

- **Characterize the record precisely.** Allegation, exhibit, testimony, order, settlement (Chapter 3, Appendix A.6). Never write "records show the company did" when the source is a complaint.
 - **Never assert intent.** Structural findings, not motives. "Common management" rather than "concealed ownership."
 - **State coverage boundaries for every negative.** So that "no records found" cannot be read as "no such thing exists."
 - **Apply the same standard to every party.** A product that is rigorous toward one side and credulous toward the other will be dismantled by whoever it disfavors, and rightly.
-

1.9 OSINT Data Feedback

1.9.1 The Most Skipped Phase

Feedback is skipped because it is never urgent, and because it exposes work that had no effect. Its absence has a specific cost: **an intelligence function without feedback cannot distinguish products that changed a decision from products that were filed unread, and it will continue producing both indefinitely.**

1.9.2 The Two Kinds of Feedback

They are different, they arrive at different times, and both are required.

Utility feedback arrives days after delivery and asks whether the product helped.

Accuracy feedback arrives weeks or months later and asks whether the assessment was right.

	Utility feedback	Accuracy feedback
Timing	3 to 10 days after delivery	After the forecast window closes
Asks	Did you read it, did it answer your question, did anything change	Did what we assessed actually happen
Source	The consumer	The world

	Utility feedback	Accuracy feedback
Improves	Format, timing, framing, requirement quality	Calibration, source weighting, analytic method
Instrument	Four questions	A scored forecast log

1.9.3 The Utility Instrument

Four questions, asked directly, one week after delivery:

1. Did you read it? If not, why not?
2. Did it answer the question you actually had?
3. Did anything change as a result: a decision, a plan, a resource, a timeline?
4. What question do you have now?

Question 3 is the one that matters. A product that changed nothing was wrong, late, obvious, aimed at the wrong person, or answering a question the consumer did not have. Those are four different diagnoses with four different remedies, and question 1 plus question 2 usually distinguish them.

Question 4 is the next requirement. Most feedback conversations generate the following cycle's PIR, which is why feedback and requirements are adjacent phases rather than opposite ends.

1.9.4 The Accuracy Record

Every assessment containing a forecast should be logged for later scoring:

Field	Content
Product ID and date	
The forecast, stated falsifiably	"X will begin regional operations by Q4"
Estimative term and probability band	"Likely, 55 to 80%"
Confidence level and its stated reason	
Resolution date	When we will know
Outcome	Occurred, did not occur, or unresolved
Post-hoc review	What we got right, what we got wrong, and which source or assumption drove the error

Calibration is the payoff. An analyst who logs forecasts and reviews them discovers, typically within a year, that their "high confidence" judgments resolve correctly at a rate they can state. That number is worth more to a consumer than any adjective, and no other habit improves analytic quality as cheaply.

Score the process, not only the outcome. A well-reasoned judgment can be wrong because the world was unlikely, and a badly reasoned one can be right by luck. Review both dimensions: was the reasoning sound given what was knowable, and was the outcome as forecast? The second without the

first teaches the wrong lessons.

1.9.5 Feedback That Changes the System

Feedback loops back into more than the next product:

Finding in feedback	System change
The consumer had a different question	Requirements process; involve the consumer earlier
The product was not read	Format, length, channel, or timing
The judgment was correct but too late	Planning: initiate slow items earlier; consider an interim product
The judgment was wrong due to a source	Source reliability rating adjusted, with the reason recorded
The judgment was wrong due to an assumption	Add that assumption class to the standing key assumptions check
A gap remained open	New collection requirement, or an honest statement that open sources cannot close it
The finding was disputed successfully	Documentation standard, characterization discipline, or coverage boundaries

1.9.6 Receiving Criticism Professionally

Lab 5 requires you to take feedback and revise. Three habits worth building now:

Separate the critique of the product from the critique of the judgment. "This was too long" and "I think you are wrong" require completely different responses.

Do not defend a judgment by adding evidence you did not use. If the assessment did not rest on it, saying so retroactively is a form of dishonesty. Revise the assessment or hold it as written.

Change the product when the critique is about the consumer's needs; hold the judgment when the critique is about the consumer's preferences. A consumer may legitimately demand a shorter product, a different format, or clearer language. A consumer asking for a more comfortable conclusion is asking for politicization, which Chapter 1 identified as the most damaging analytic failure, and the answer is a respectful no.

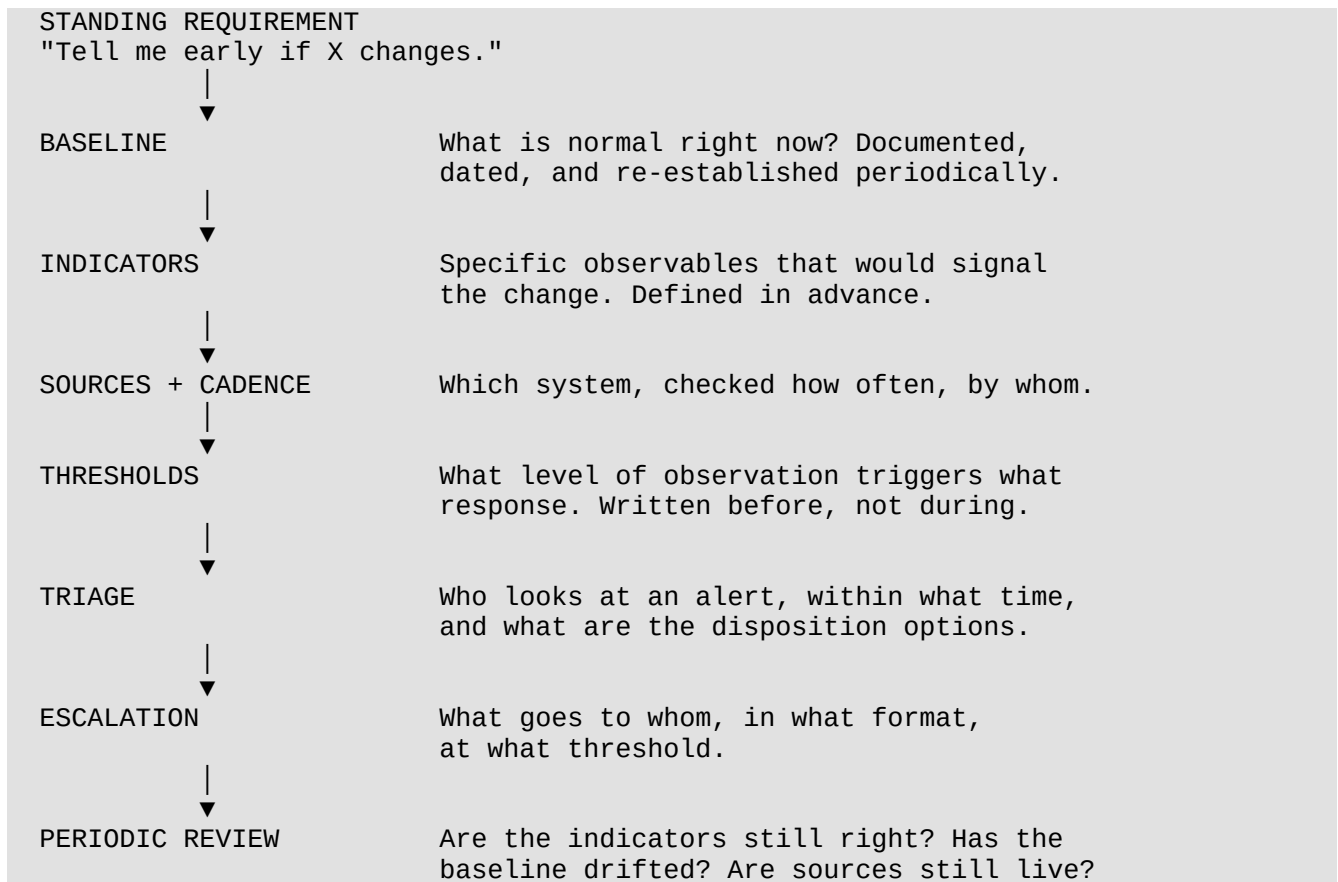
1.10 Continuous Collection of OSINT Data

1.10.1 Investigations End; Monitoring Does Not

An investigation is bounded by a decision and a deadline. Continuous collection is structured around a standing question and runs until the question stops mattering. The difference is architectural, not just temporal.

	Episodic investigation	Continuous collection
Trigger	A decision	A standing concern
Requirement	A PIR with a deadline	A standing requirement with a review cadence
Success	The question is answered	Change is detected early enough to act
Collection	Broad and deep, once	Narrow and repeated, on a schedule
Analysis	Integrative, once	Comparison against a baseline
Product	A report	An alert, plus a periodic summary
Main failure	Missing the answer	Alert fatigue, silent breakage, and baseline drift

1.10.2 The Architecture



1.10.3 Designing Indicators and Thresholds

An indicator is only useful if it is observable, attributable, and discriminating.

Property	Test
Observable	Can it actually be seen in a source we have access to?
Attributable	If we see it, do we know it relates to our subject?
Discriminating	Does its presence distinguish between the states we care about, or is it consistent with both?
Timely	Would we see it early enough to act?
Sustainable	Can we check it at this cadence indefinitely?

Thresholds convert observation into action. A three-tier structure is usually enough:

Tier	Condition	Action	Who acts
Watch	One indicator observed	Log, note, no escalation	Analyst
Elevated	Two or more indicators, or one high-weight indicator	Short written note to the named consumer within 24 hours	Analyst, informing lead
Trigger	Threshold set of indicators, or a decisive single observation	Flash product plus a pre-agreed action	Lead, informing decision maker

Writing the thresholds in advance is the entire value. In the moment, everything looks either urgent or explicable, and a pre-agreed threshold is what prevents both over-reaction and rationalization.

1.10.4 The Three Failure Modes of Continuous Collection

Alert fatigue. Too many low-value alerts train the recipient to ignore all of them, including the one that mattered. Remedies: tune thresholds against observed false positive rates, aggregate low-tier observations into a periodic summary rather than alerting individually, and require every alert type to have a defined action. **An alert with no defined action should be deleted, not tuned.**

Silent breakage. A source changes its structure, a feed stops, an account is suspended, a portal moves. Collection stops and nobody notices, and weeks later an analyst concludes that activity ceased. Remedies: monitor for the absence of expected input, not only for its presence; test each source on a schedule; and record when each source last returned something.

Baseline drift. Normal changes gradually, the baseline is never re-established, and the monitoring slowly measures the wrong thing. Remedy: a scheduled baseline refresh, with the date recorded.

1.10.5 The Review Cadence

Interval	Activity
Daily or per-shift	Triage alerts; dispose of each with a recorded decision

Interval	Activity
Weekly	Roll low-tier observations into a short summary; confirm every source returned something
Monthly	Review false positive and false negative rates by indicator; tune thresholds
Quarterly	Re-validate indicators against the standing requirement; refresh the baseline; confirm the consumer still needs this
Annually	Re-derive the standing requirement from the current decision environment; retire what no longer matters

That last row is the one organizations skip, and the result is monitoring programs that outlive the concerns that created them, consuming attention indefinitely. **Retiring a monitoring requirement is a legitimate and valuable act**, and it should be documented as deliberately as creating one.

1.10.6 Continuous Collection and Ethics

Two constraints intensify when collection becomes ongoing:

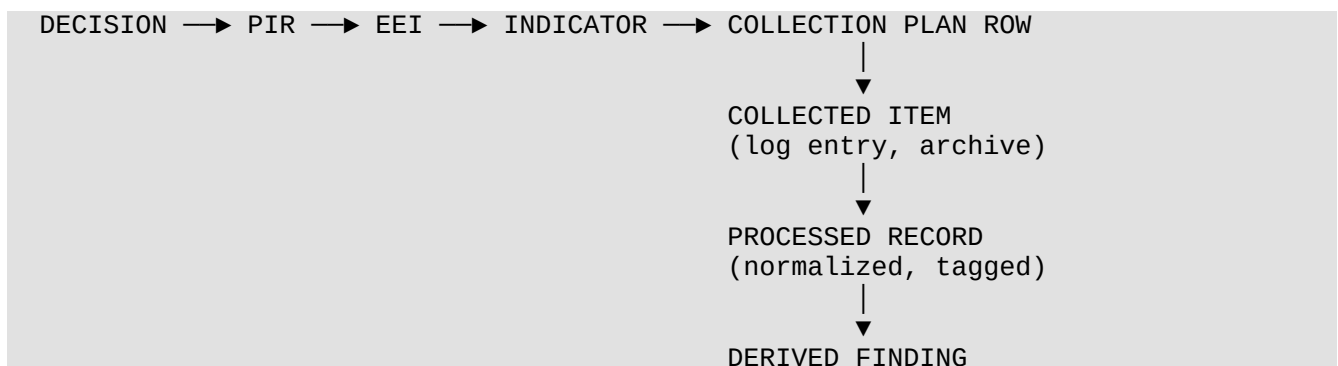
Retention. Episodic collection produces a bounded record. Continuous collection accumulates indefinitely, and accumulation about individuals is where the mosaic effect (Chapter 2) becomes acute. Every monitoring program needs a written retention limit and a purge schedule that actually runs.

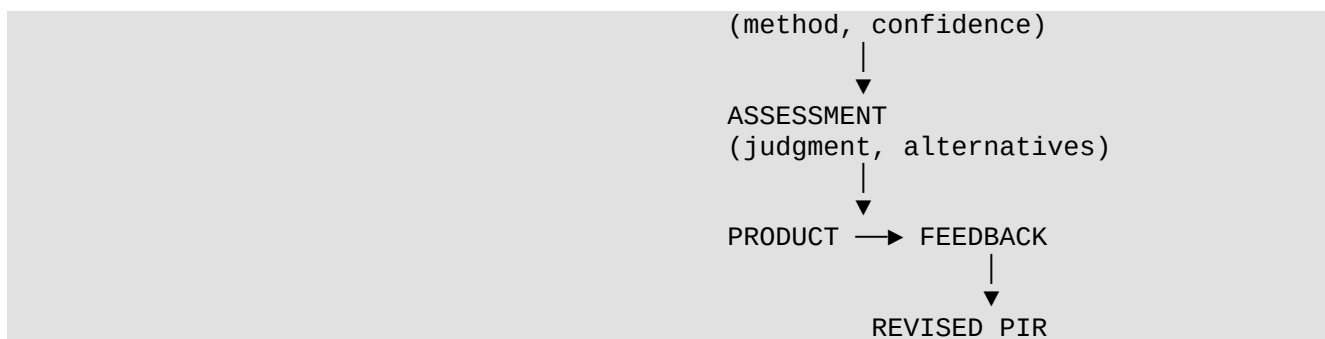
Scope creep. A monitoring program built for one purpose gradually collects adjacent material because it is easy. The quarterly review must include the question: is everything we are collecting still tied to the standing requirement? Anything that is not gets dropped, and the drop gets recorded.

1.11 Cross-Cutting: Documentation, Ethics, and Quality Control

1.11.1 The Documentation Chain

The methodology produces a chain in which every artifact traces to the one before it. This is what makes the whole thing auditable.





The audit test. Pick any sentence in a finished product. You should be able to walk backward: which assessment, from which derived findings, from which items, collected under which plan row, against which indicator, serving which EEI, under which PIR, supporting which decision. If any link is missing, the sentence is not defensible.

1.11.2 The Ethics Annex

Every product in this chapter carries one. Its contents:

- Every individual who appeared anywhere in the work, and in what capacity.
- What was deliberately not collected, and why.
- Any material encountered that was not lawfully public, and how it was handled.
- Any offer of material from a third party, and the disposition.
- The retention decision for the collected material.

Why it is a positive document, not a disclaimer. An ethics annex that records what you declined to do is evidence of professional discipline, and it is the document that protects both the subject and you.

1.11.3 Quality Control Checkpoints

Checkpoint	Who	When
Requirement review	Consumer	Before planning
Plan review, including legal	Lead or peer	Before collection
Collection log spot check	Peer	Midway through collection
Processing gate	Self, against the Section 1.5.6 list	Before exploitation
Derived findings review	Peer, on the highest-impact derivations	Before analysis
Analytic review	Peer, focused on alternatives and confidence	Before dissemination
Characterization check	Self, against Chapter 3 Appendix A.6	Before dissemination
Post-delivery review	Consumer and team	After feedback

For solo work, the peer reviews become self-reviews performed after a deliberate break, using a written checklist rather than a reread. It is a weaker control, and it should be named as one in the product's limitations.

Part 2: Comparison Tables

2.1 Information vs. Intelligence

Dimension	Information	Intelligence
Origin	Exists whether or not anyone needs it	Collected against an endorsed requirement
Driver	Availability	A decision maker's question
Evaluation	Usually none	Source reliability and claim credibility, assessed separately
Processing	Whatever form it arrived in	Normalized, deduplicated, entity-resolved, indexed
Exploitation	None	Derived findings extracted, with method and confidence
Integration	Isolated	Compared across items, categories, and time
Judgment	None	Explicit, with a confidence level and its stated reason
Alternatives	Not considered	Written and tested by disconfirmation
Uncertainty	Unstated	Assumptions and ranked gaps named
Audience	Anyone or no one	A named consumer, in a matched format
Timing	Whenever	Before the decision, with room to act
Accountability	None	A named analyst owns the judgment and can be scored later
Traceability	None	Every sentence walks back to an item, a plan row, and a requirement
Volume	Grows without limit	Deliberately bounded by the requirement
Test of quality	Is it accurate?	Did a decision change for the better?

Dimension	Information	Intelligence
Failure mode	Missing or wrong	Late, unread, unactionable, or hedged into uselessness

2.2 Collection vs. Processing vs. Exploitation vs. Analysis

Dimension	Collection	Processing	Exploitation	Analysis
Operates on	The world	The form of collected items	The content of a single item	The set of all findings
Question	Can I obtain it?	Can I sort and join this?	What does this item contain that I could not see?	What does all of it mean?
Output	Raw items plus a log	Normalized, tagged, indexed dataset	Derived findings with method and confidence	A judgment with confidence, alternatives, assumptions, gaps
Judgment applied	Source selection and lawfulness	Almost none; consistency decisions only	Substantial, item by item	Substantial, across items
Typical operations	Query, retrieve, archive, log	Normalize, deduplicate, resolve entities, translate, transcribe, OCR, tag	Geolocate, chronolocate, read metadata, interpret imagery, extract entities and cross-references	Hypothesize, disconfirm, weigh, assign confidence, rank gaps
Skill emphasis	Knowing where things live	Discipline and consistency	Technique and care	Reasoning and calibration
Time share (typical)	30%	20%	15%	15%
Failure mode	Wrong entity, wrong jurisdiction, no boundary, no archive	Mixed time zones, split or merged entities, uncollapsed duplicates	Confident wrong derivation; interpretation recorded as observation	Confirmation bias, layering, uncalibrated confidence
Consequence of skipping	Nothing to work with	Nothing comparable; false timelines	Value left in the items; findings undefendable	A pile of facts and no answer
Who can do it	Anyone following a plan	Anyone disciplined	Trained practitioner	Trained analyst, accountable
Quality gate	Every item logged and archived; boundaries	Dates, names, entities normalized;	Findings recorded separately, with method and	Alternatives tested; assumptions written; confidence

Dimension	Collection	Processing	Exploitation	Analysis
	recorded	duplicates marked	confidence	justified

2.3 Intelligence Requirements vs. Collection Requirements

Dimension	Intelligence requirement	Collection requirement
Answers	What does the decision maker need to know?	What must a collector go get?
Phrased as	A question about the world	A specific, retrievable item from a named source
Level	PIR and EEI	Indicator and plan row
Owner	Consumer, endorsed	Collector, tasked
Written by	Analyst, with the consumer	Analyst, for the collector
Test	Would the answer change a decision?	Could a stranger execute it and know when they are done?
Scope	Bounded by the decision	Bounded by the source and date range
Time frame	Tied to the decision date	Tied to collection latency
Example	"Will Competitor X begin regional operations within 18 months?"	"Search the recorder's grantor/grantee index in these 4 counties for these 8 name variants, 24-month range"
Changes when	The decision changes	The source landscape or the indicator set changes
Failure if wrong	Effort answers a question nobody had	Collector cannot execute, or collects the wrong thing
Documented in	The requirement record	The collection plan
Reviewed by	The consumer	The lead, plus a legal and ethical check per row

2.4 Manual vs. Automated Collection

Dimension	Manual	Automated
Throughput	Low	High
Per-item judgment	Applied every time	Only what was encoded in advance
Best suited to	Dockets, filings, imagery interpretation, ambiguous	Monitoring, alerting, repeated checks, bulk retrieval where

Dimension	Manual	Automated
	identity work	permitted
Setup cost	None	Real: build, test, maintain
Marginal cost	High per item	Near zero per item
Consistency	Variable; fatigue and drift	Perfectly consistent, including consistently wrong
Error signature	Random, individual, often caught by the analyst	Systematic, replicated at scale, often invisible
Documentation	Requires manual discipline	Logged by construction, which is a genuine advantage
Reproducibility	Depends on the log	High, if the script and parameters are retained
Legal exposure	Low	Higher: terms of service, rate limits, computer misuse questions
Detectability by the subject	Low	Higher: volume and pattern are visible
Failure when it breaks	Analyst notices	Silent; requires a heartbeat check for absence of input
Right use in continuous collection	Triage and disposition of alerts	Generating the alerts
Right use in this course	All labs	Only where permitted, disclosed, and rate-limited (Rule 5)

2.5 Tactical vs. Strategic Reporting

Dimension	Tactical reporting	Strategic reporting
Consumer	Operator, responder, field staff, analyst on shift	Executive, board, policy maker, planner
Time horizon	Seconds to days	Months to years
Purpose	Enable an immediate action	Shape direction, posture, and resource allocation
Length	One paragraph to two pages	Five to thirty pages, or a structured briefing
Specificity	Names exact things: addresses, identifiers, times	Deals in trends, categories, ranges, and scenarios
Perishability	High; may expire within hours	Low; useful for months or years
Confidence handling	Stated, but converted into an actionable default ("treat as	Stated with full reasoning and alternatives

Dimension	Tactical reporting	Strategic reporting
	unconfirmed, act accordingly")	
Alternatives shown	Rarely; the operator needs a call	Always; usually as scenarios with discriminating indicators
Sourcing shown	Minimal; a reference to the log	Extensive; the product is expected to be challenged
Delivery	Pushed, in the channel they are already watching	Scheduled, briefed, discussed
Typical products	Flash report, bulletin, indicator list, watch note	Assessment, scenario set, annual threat picture, program review
Feedback loop	Immediate and informal, in the same channel	Formal, delayed, tied to the decision cycle
Failure mode	Arrives late or in the wrong format	Unread, or read and not acted on
Relationship	Feeds observations upward	Sets the requirements that drive tactical watching

2.6 Common Collection Errors and Their Impact

Error	Phase where it originates	Immediate effect	Downstream impact	Prevention
Requirement not answerable	Requirements	Collection has no target	Entire effort produces material nobody needed	Four tests before planning
Requirement not endorsed	Requirements	Consumer disowns the product	Work discarded at delivery	Written endorsement before planning
Wrong entity searched	Collection	Nothing found under the given name	Material findings under former names never surface; false clean result	Entity expansion before any search
Wrong jurisdiction	Collection	Empty result	False negative reported as a finding	Jurisdiction determination before records work
No coverage boundary	Collection	"Nothing found"	A limitation is read as a conclusion	Negative finding template, every time
No archiving	Collection	Live URL cited	Finding evaporates when the page	Archive at collection

Error	Phase where it originates	Immediate effect	Downstream impact	Prevention
			changes	
Retroactive logging	Collection	Log assembled at the end	Timestamps invented; work unauditable	Log at the moment of collection
Collection drift	Collection	Time spent on an interesting thread	Requirement still open at deadline	Plan rows and a stopping condition
Over-collection	Collection	Thousands of items	Processing and analysis swamped; triage discards blindly	Saturation test; emphasis matrix
Mixed time zones	Processing	Timeline looks fine	Events sequenced wrongly; false causation asserted	Normalize to one standard, record zones
Entity split	Processing	One actor appears as three	Pattern invisible; activity under-counted	Canonical entity list with variants
Entity merge	Processing	Three actors appear as one	Relationship invented; conspiracy manufactured	Require a second identifier before merging
Duplicates uncollapsed	Processing	Item count looks strong	Corroboration inflated; one wire story counted five times	Deduplicate and mark, do not delete
Interpretation recorded as observation	Exploitation	Reads like a fact	Assessment propagates as evidence; product collapses under challenge	Two-column discipline
Cross-references not chased	Exploitation	Documents read in isolation	Most of the available value left uncollected	Every cross-reference becomes a plan row
Single-category corroboration	Analysis	Feels confirmed	Circular reporting mistaken for confirmation	Count distinct categories, not items
Alternatives not tested	Analysis	A clean story	Confirmation bias; the obvious explanation wins by default	ACH, with disconfirmation scoring
Confidence uncalibrated	Analysis	Confident language	Consumer over-relies; credibility	State the reason for the level

Error	Phase where it originates	Immediate effect	Downstream impact	Prevention
			lost when wrong	
Wrong format or channel	Dissemination	Product delivered	Not read; effort wasted	Match product to consumer; confirm receipt
No feedback collected	Feedback	Nothing visible	Function cannot improve or demonstrate value	Four questions, one week later
Alert with no defined action	Continuous	Noise	Alert fatigue; the important alert is ignored	Every alert type has a defined action or is deleted
Silent collection breakage	Continuous	Quiet	Absence of activity inferred from absence of collection	Heartbeat check for missing input

2.7 Benefits and Challenges of Each Lifecycle Phase

Phase	Benefit when done well	Challenge	Characteristic failure	Cost of skipping
Requirements	Everything downstream has a purpose and a stopping point	Consumers rarely arrive with an answerable question; requires negotiation	"Tell me everything about X"	The whole effort answers a question nobody had
Planning	Executable tasking; legal exposure resolved before it exists; slow items started early	Feels like delay when the deadline is close	A list of websites instead of a plan	Drift, duplication, legal exposure, missed slow items
Collection	A defensible, reproducible evidence base	Discipline feels like overhead; volume grows fast	Retroactive logging and unbounded negatives	Findings that cannot be defended or reproduced
Processing	Material becomes comparable; timelines become possible	Invisible work with no visible product; consistently under-budgeted	Skipped, or done partially at the end	False timelines, split entities, inflated corroboration
Exploitation	Value extracted that was not on the face of the items	Technique-heavy and error-prone; easy to rush	Interpretation recorded as observation	Most of the collected value never surfaces

Phase	Benefit when done well	Challenge	Characteristic failure	Cost of skipping
Analysis	A judgment a consumer can act on, with honest uncertainty	Requires actively seeking to be wrong	The first explanation wins	A pile of facts and no answer
Dissemination	The work reaches the decision in a usable form	Format and timing decisions feel secondary to content	Right product, wrong length, wrong channel, too late	Correct work with zero effect
Feedback	The function learns; requirements improve; calibration develops	Never urgent; exposes work that had no effect	Skipped entirely	Cannot distinguish useful products from unread ones
Continuous	Change is detected early enough to act	Fatigue, silent breakage, baseline drift, scope creep	Alerts nobody reads	The next event is a surprise, again

Part 3: Case Studies

Each case walks the full methodology, phase by phase, and states the decisions made at each gate. Cases marked *historical* describe real documented events in general terms. Cases marked *composite* are built from documented professional practice and are not real organizations.

3.1 Cybersecurity: A Credential Phishing Campaign Against a University System

(*composite*)

Situation. A four-campus university system detects a small number of successful account compromises. The security team has 6 people and no intelligence function. The interim CISO needs to know whether these are isolated or part of a campaign, and what to do this week.

Phase	Decisions and actions	Output
Requirements	Consumer: interim CISO, decision due in 5 days on whether to force a system-wide credential reset (disruptive, mid-term) or a targeted one. PIR:	Endorsed PIR, 3 EEIs, 11 indicators

Phase	Decisions and actions	Output
	"Are the observed compromises part of a coordinated campaign targeting this institution, and is the campaign ongoing?" Three EEIs: infrastructure, targeting scope, and current activity	
Planning	Emphasis matrix put certificate transparency, domain registration, and passive DNS in the quick-win quadrant; scan data and archived captures in the core quadrant. Legal review: all rows public and passing the four tests. OPSEC: research browser, no interaction with any suspected infrastructure. Prohibited list included any attempt to access the phishing sites' back end	11-row collection plan, prohibited-activity statement
Collection	Entity expansion first: enumerated the institution's legitimate domains, subdomains, and address ranges. Then certificate transparency for brand-similar names, registration data, passive DNS, scan data on owned space, and archived captures of the phishing pages where indexed. Negative results logged with boundaries	41 items, all archived and logged
Processing	Normalized all timestamps to UTC; resolved domain variants to a canonical list; deduplicated three certificate entries that described the same issuance; tagged every item to an indicator	Structured dataset, master timeline
Exploitation	Extracted registrant patterns, name server assignments, hosting addresses, and issuance timing. Derived finding: 11 of 14 domains shared 2 name servers and were registered in 3 tight clusters. Separately: archived captures of two pages showed identical page structure and a common typographical error	9 derived findings, each with method and confidence

Phase	Decisions and actions	Output
Analysis	Hypotheses: (a) one coordinated campaign, (b) several unrelated opportunistic actors, (c) a commodity phishing kit used independently by many. Discriminating evidence: the shared typographical error plus the name server overlap plus the registration clustering. (c) could not be fully excluded, which capped confidence	Assessment: coordinated campaign, moderate to high confidence; ongoing, likely
Dissemination	Flash note (4 sentences) to the CISO within 3 hours of the key finding; one-page brief the next morning; a structured indicator table to the detection engineers in their channel; nothing to the wider IT list, to avoid tipping	3 products, matched to 3 consumers
Feedback	Utility feedback at 1 week: the CISO chose a targeted reset scoped by the identified targeting pattern rather than a system-wide reset. Question 4 produced the next PIR: "which of our services can be accessed without phishing-resistant authentication?"	Decision changed; new requirement
Continuous	Standing requirement established with 4 indicators, daily certificate transparency checks for brand-similar issuance, a defined 3-tier threshold, and a monthly false positive review	Monitoring plan with owners and cadence

What the methodology contributed. The decision that changed was scope: a targeted reset instead of a system-wide one, which the CISO estimated saved several thousand hours of institutional disruption. Note that the analysis did *not* reach high confidence, and the recommendation was still actionable. **A moderate-confidence judgment with a clearly stated basis is more useful than a confident one that cannot be defended.**

3.2 Criminal Investigation: Illegal Dumping in a Rural County

(composite)

Situation. A county environmental enforcement office has 60 illegal dumping incidents over 18 months across a rural area, and two officers. Individual citations have not reduced the rate.

Phase	Decisions and actions	Output
Requirements	Consumer: enforcement supervisor. Decision: whether to keep responding site by site or to build a case against a hauling operation. PIR: "Is the dumping being generated by a small number of commercial haulers, and if so, which?"	PIR, 4 EEIs
Planning	Category-first plan: incident records (internal), waste hauler licensing and transfer station records (government records), corporate filings, satellite and aerial imagery, court records, and legal notices. Legal note: all public. Individuals policy written: haulers appear as business entities; no research on residents near the sites	16-row plan
Collection	Structured internal incident data first, then licensing records, then imagery for the 11 highest-volume sites. Records request filed on day one for transfer station tonnage logs, which had 3-week latency	74 items; one slow item in flight
Processing	Incident reports structured into a table with normalized dates, coordinates, material type, and volume. Hauler names resolved to canonical entities across 3 spelling variants and 2 former names	Structured incident dataset, entity list, timeline
Exploitation	Material composition exploited from photographs: construction debris of a type associated with a specific demolition trade appeared at 38 of 60 sites. Imagery exploitation established that 9 sites were on 3 parcels with a single access road. Licensing records exploited for	Derived findings: material signature, geographic corridor, entity relationships

Phase	Decisions and actions	Output
	cross-references to bonding and insurance filings, which named additional entities	
Analysis	Pattern analysis showed dumping clustered by day of week and by corridor. Alternative hypotheses tested: many small offenders, one commercial hauler, or a transfer station pricing effect pushing haulers to dump. The transfer station tonnage logs, when they arrived, discriminated: a licensed hauler's declared tonnage dropped 60% while its permit activity continued	Assessment: a small number of commercial haulers, high confidence for one entity, moderate for a second
Dissemination	Operational assessment to the supervisor and the county attorney, with the evidentiary characterization annex; a tactical watch note to field officers naming the corridor and the days of week; nothing public, to avoid alerting	2 products
Feedback	The supervisor redirected patrol to the corridor and the county attorney opened a case. Accuracy feedback at 90 days: the forecast that dumping would concentrate in the corridor held; the second hauler was not implicated, and the source that had suggested it was downgraded	Accuracy record; source rating adjusted
Continuous	Standing monitoring of licensing status and transfer station declared tonnage, quarterly, with a threshold at a 30% unexplained tonnage drop	Monitoring plan

Lesson. The decisive item was a **slow item initiated on day one**. Had the records request gone out in week three, it would have arrived after the decision. Planning that front-loads latency is not administrative tidiness; it is what makes the discriminating evidence available in time.

3.3 Competitive Intelligence: A Health System Assessing a Rival's Ambulatory Expansion

(composite)

Situation. A regional health system must decide within one quarter whether to accelerate a \$40M ambulatory build-out, ahead of a suspected competitor expansion.

Phase	Decisions and actions	Output
Requirements	Consumer: chief strategy officer, decision at the Q3 board. PIR: "Will the competitor open ambulatory sites in our three priority submarkets within 24 months?" Answerability assessment stated up front that intent is not reachable from open sources, so the requirement was framed around necessary observable precursors	PIR, 5 EEIs, 17 indicators
Planning	Emphasis matrix: certificate-of-need and licensing filings, property records, and permits as core effort; job postings and archived captures as quick wins; imagery as core; podcasts and investor calls as opportunistic. Ethics: competitor employees appear only via public professional roles; no contact, no elicitation	17-row plan, prohibited-activity statement
Collection	State health facility licensing and regulatory filings, county property and permit records in 3 submarkets, corporate registrations for affiliated entities, commercial imagery over 4 candidate parcels, job postings by metro and role family, archived captures of the competitor's locations page, and one industry podcast appearance by a competitor executive	96 items
Processing	Normalized submarket geography to county and ZIP; resolved 7 affiliated entity names to 2 canonical parents; separated	Structured dataset, timeline with precision column

Phase	Decisions and actions	Output
	event dates from publication dates throughout; built a master timeline at month precision	
Exploitation	Regulatory filings exploited for square footage, service lines, and projected opening dates. Imagery exploitation established site preparation on 1 of 4 parcels. Job posting exploitation showed clinical roles appearing 5 to 7 months before each of the competitor's prior openings, from historical comparison. Podcast transcript exploitation captured an executive statement about a financing structure with a closing date	Derived findings including a 5-to-7-month hiring lead indicator
Analysis	Hypotheses: full expansion into all three submarkets, a single-site test, or a filing made to preserve optionality without intent to build. Discriminating evidence: site preparation visible in imagery on one parcel, plus regulatory filings on two, plus absence of clinical hiring in the third. Confidence differed by submarket, and the product said so	Assessment: submarket A likely within 12 months (high confidence), B possible within 24 (moderate), C not supported (moderate)
Dissemination	One-page brief to the CSO; a two-page assessment for the board packet with the sourcing annex; a watch note to the regional directors with the hiring lead indicator so they could see it themselves	3 products
Feedback	The board accelerated in submarket A only, rather than all three, which preserved roughly \$26M for other uses pending further evidence. Question 4 generated the next PIR about payer contracting	Decision changed; scope narrowed
Continuous	Standing monitoring on regulatory filings (weekly), job	Monitoring plan with a defined trigger

Phase	Decisions and actions	Output
	postings by metro (weekly, aggregated monthly), and imagery over 4 parcels (quarterly), with the hiring lead as the primary trigger indicator	

Lesson. Differentiated confidence by submarket is what made the product actionable. A single blended judgment ("expansion likely") would have led to a \$40M commitment across three markets. **Precision about where you are and are not confident is often worth more than the judgment itself.**

3.4 Disaster Response: Earthquake Damage Assessment

(historical pattern, described in general terms)

Situation. Following the February 2023 earthquakes in Türkiye and Syria, response organizations required rapid situational awareness across a very large affected area with damaged infrastructure and limited access.

Phase	How it ran
Requirements	Multiple consumers with different questions: which settlements have the highest structural damage, which routes are passable, where are populations concentrated, and where is aid not reaching. Requirements were set by responder need rather than by data availability
Planning	Source planning drew on satellite imagery providers activating humanitarian access programs, volunteer mapping communities, social and broadcast reporting, and official announcements. Latency was the binding constraint, so imagery tasking and volunteer mobilization were initiated immediately
Collection	Post-event satellite and aerial imagery; participant photographs and video; official statements; volunteer-contributed map data; broadcast reporting
Processing	The dominant workload. Enormous volumes of unstructured reports required geolocation, deduplication, translation, and categorization before any of it could direct a responder. Volunteer capacity was applied here more than anywhere else

Phase	How it ran
Exploitation	Damage assessment from imagery comparison against pre-event baselines; geolocation of participant material; road passability derived from imagery and reports; building-level damage classification in some areas
Analysis	Aggregation into damage severity by settlement, comparison against population data, and identification of areas with high assessed damage and low reported aid presence, which is the analytic product responders actually needed
Dissemination	Map layers and structured datasets in formats usable in the field on intermittent connectivity, updated continuously, plus periodic written summaries
Feedback	Continuous and informal from field users; several later reviews identified coordination with official structures and duplication between volunteer efforts as the recurring weaknesses
Continuous	Monitoring continued through the recovery phase, with the requirement shifting from damage assessment to shelter, services, and displacement tracking

Methodology lessons. Three, all of which recur in disaster work.

Processing is the bottleneck, not collection. The constraint was never a shortage of reports. It was the capacity to geolocate, deduplicate, translate, and categorize them fast enough to be useful.

Requirements must be set by responder need. Producing what is easy to produce, rather than what a responder can act on, is the characteristic failure of enthusiastic volunteer effort.

Reporting bias under-represents the most affected. Areas with destroyed connectivity generate the fewest reports, which is the availability bias from Chapter 2 in its most consequential form. Products must carry that limitation prominently, or they will direct resources toward the areas best able to ask for them.

3.5 Journalism: A Newsroom Visual Investigation

(historical pattern, described in general terms)

Situation. Major newsrooms now operate dedicated visual investigation units that reconstruct contested events from open material. The pattern below reflects the published methodology of that work.

Phase	How it runs
Requirements	The editorial question is the requirement, and it is deliberately narrow: not "what happened" but "did a specific weapon type strike a specific location at a specific time," or "was the official account of a sequence accurate." Narrow questions are what make verification tractable
Planning	Source planning is explicit: participant video, commercial satellite imagery, weather and solar data, official statements, prior imagery for geolocation reference. Legal review addresses jurisdiction, privacy, and the treatment of identifiable individuals in footage
Collection	Systematic collection of every available angle, with archiving at collection because material is routinely deleted. Provenance recorded for each item: who published first, when, and on what basis
Processing	Items normalized to a single timeline, duplicates and re-uploads collapsed, and translation applied where needed. The timeline is the central artifact
Exploitation	The technical heart of the work: geolocation from fixed features, chronolocation from shadow geometry and weather records, munition or vehicle identification from visual characteristics, and blast or ballistic analysis where applicable. Every derived finding records the features matched
Analysis	Reconstruction of the sequence, tested against alternative explanations, with the specific claims separated by confidence. Independent expert review is routinely commissioned for technical derivations
Dissemination	Publication with the evidence shown: coordinates, source imagery, and the reasoning, so that any reader can check the work. This is a verification strategy as much as a transparency one
Feedback	Adversarial: interested parties attempt to discredit the work publicly. Surviving that is the quality test. Corrections are issued visibly when the challenge succeeds
Continuous	Ongoing monitoring of a conflict or subject, with the same techniques applied repeatedly and a growing reference library of geolocated features

Methodology lessons.

A narrow requirement is what makes verification possible. Broad questions cannot be verified; specific claims can.

Publishing the evidence is a quality control mechanism. Inviting adversarial checking is uncomfortable and it is the reason this work has become authoritative.

Exploitation is where the expertise lives. Anyone can collect the videos. The geolocation and chronolocation are the skilled labor, and they are what the product actually rests on.

3.6 Fraud Detection: A Post-Disaster Contractor Scheme

(composite)

Situation. After a severe hailstorm, a state consumer protection unit receives 140 complaints about roofing contractors within six weeks. Individual complaint handling is not keeping pace.

Phase	Decisions and actions	Output
Requirements	Consumer: unit supervisor. Decision: whether to pursue individual complaints or build a pattern case. PIR: "Do the complaints reflect a small number of connected operations, and what is the operating pattern?"	PIR, 4 EEIs
Planning	Plan drew on complaint records (internal), contractor licensing, corporate filings, county permit records, domain registration and archived captures, legal notices, and court records. Individuals policy: complainants are private individuals whose details are used only in aggregate and never appear in any product	19-row plan
Collection	Complaint records structured first; then licensing status for each named business; corporate registry searches on entity and officer names; permit records for the addresses involved; registration data and archived captures for the businesses'	210 items

Phase	Decisions and actions	Output
	websites	
Processing	Business names resolved to canonical entities across 31 observed variants; complainant data aggregated and de-identified immediately at processing, before any analysis; complaint dates, contract dates, and payment dates normalized separately	Structured dataset; de-identified complaint aggregate
Exploitation	Registration data exploited: 9 business websites registered within an 11-day window, sharing 2 name servers. Archived captures exploited: 6 sites shared identical policy text with one company's name left uncorrected. Corporate filings exploited by officer name: 3 entities shared a manager. Permit records exploited: 88% of the addresses had no permit filed for the work performed	Derived findings across 4 categories
Analysis	Hypotheses: independent opportunistic operators drawn by the storm, one connected operation using multiple names, or a franchise or lead-generation arrangement producing surface similarity without common control. The shared manager plus the uncorrected policy text plus the registration clustering discriminated in favor of the second, though a lead-generation arrangement could not be fully excluded for 3 of the 9	Assessment: a connected operation covering at least 6 of 9 businesses, high confidence; 3 unresolved
Dissemination	Investigative memo to the supervisor and counsel with full characterization discipline (no allegation of fraud, only structural findings and permit non-compliance); a public consumer advisory describing the <i>pattern</i> without naming	3 products, 3 very different audiences

Phase	Decisions and actions	Output
	unadjudicated businesses; a referral package to the licensing board	
Feedback	The unit consolidated 140 complaints into one pattern investigation. Feedback from counsel drove a revision: the first draft used the word "scheme," which was replaced throughout with descriptions of the structural findings	Revision; wording standard adopted
Continuous	Standing post-disaster monitoring: after any declared weather event, automatic checks of new domain registrations containing storm and roofing terms in the affected counties, plus new contractor registrations, with a defined threshold	Monitoring playbook

Lesson. The de-identification decision at **processing**, not at dissemination, is what kept the effort clean. Once complainant details are carried into analysis they tend to appear in products. Handling privacy at the earliest possible phase is both easier and safer than filtering it at the end.

3.7 National Security: Open Source Warning Before a Military Invasion

(historical, introduced in Chapter 2 and revisited here as a methodology case)

Situation. Through late 2021 and early 2022, military buildup around Ukraine was extensively documented in open sources by civilian analysts, commercial imagery providers, and journalists, alongside government assessments.

Phase	How it ran
Requirements	For civilian analysts, the requirement was self-set and narrow: track the location and composition of identifiable units over time. For governments, the requirement was tied to policy decisions about warning, deterrence, and allied preparation
Planning	Source planning centered on commercial satellite imagery, publicly posted video of rail and road movements, unit insignia visible in imagery, and open transport and logistics reporting. The plan was inherently continuous rather than episodic

Phase	How it ran
Collection	Sustained daily collection over months, with archiving at collection because social material was routinely deleted
Processing	Massive normalization workload: geolocating thousands of clips, resolving unit identities across variant references, deduplicating re-uploads, and maintaining a running timeline and order-of-battle dataset
Exploitation	Geolocation of movement footage, chronolocation, unit identification from insignia and vehicle markings, and imagery comparison against baseline garrison locations. Each derived finding was published with its matched features
Analysis	Aggregation into unit-level disposition over time; comparison against the historical pattern of exercises versus preparation; explicit alternative hypotheses (exercise, coercive signaling, invasion preparation) with attention to the indicators that discriminate, notably field hospitals, blood supply, logistics density, and the movement of units from distant districts
Dissemination	Two channels operating differently: civilian analysts published continuously in public, and governments made an unusual choice to disclose assessments publicly in advance, partly because open source corroboration meant those disclosures could be independently checked
Feedback	Post-event review across the analytic community examined which indicators had proven discriminating and which had not; the exercise-versus-preparation discrimination problem became a widely studied case
Continuous	The monitoring architecture built during this period continued and became a permanent capability for many organizations

Methodology lessons.

The discriminating indicators were logistics, not combat power. Force concentration was consistent with both hypotheses. Medical, fuel, and sustainment indicators discriminated. This is Step 4 of the analytic sequence in its purest form.

Shareability changed what the intelligence could do. Chapter 2 made this point structurally; here it is operational. Because the assessment could be corroborated from open material, public disclosure was a

usable option, and that option shaped allied coordination.

Continuous collection was the enabling structure. No episodic effort produces a months-long order-of-battle picture. The capability existed because the monitoring architecture was built and sustained.

3.8 Cross-Case Comparison

Case	Requirement type	Decisive phase	Discriminating evidence	Decision changed
University phishing	Episodic, 5-day deadline	Exploitation (infrastructure clustering)	Shared typographical error plus name server overlap	Targeted reset instead of system-wide
Illegal dumping	Episodic, feeding a case	Planning (slow item started day one)	Transfer station tonnage decline	Pattern case instead of site-by-site citations
Ambulatory expansion	Episodic, quarterly decision	Analysis (differentiated confidence)	Imagery site prep plus hiring lead indicator	Accelerated in one submarket, not three
Earthquake response	Continuous, multi-consumer	Processing (triage at volume)	Damage severity against aid presence	Resource direction to under-served areas
Newsroom investigation	Episodic, narrow claim	Exploitation (geolocation, chronolocation)	Matched fixed features and shadow geometry	Public record of a contested event
Contractor fraud	Episodic, becoming continuous	Processing (de-identification) and exploitation	Shared manager, uncorrected policy text, registration clustering	140 complaints consolidated into one case
Invasion warning	Continuous	Analysis (indicator discrimination)	Logistics and medical indicators	Public pre-emptive disclosure; allied preparation

Two patterns run through all seven. **The decisive phase is almost never collection.** In six of seven, the phase that determined the outcome was planning, processing, exploitation, or analysis. And **every case turned on a small number of discriminating items**, not on the volume of material collected. Both findings should change how you allocate your effort.

Part 4: Chapter Summary

A methodology exists to make work repeatable, auditable, efficient, and diagnosable. The fourth

reason is the one practitioners underrate: when an effort fails, named phases let you locate the break. Without them, every failure looks like a collection failure, and the organization responds by collecting more, which is almost always the wrong remedy.

The methodology has nine phases: requirements, planning, collection, processing, exploitation, analysis, dissemination, feedback, and continuous collection. Each has an output and a quality gate. The clean sequence is a teaching model; real practice runs phases in parallel, jumps urgent findings forward, loops backward when exploitation generates new requirements, and revises the requirement mid-effort when the decision changes. What you must never do is compress by deleting a phase, because a deleted phase moves its cost downstream where it is larger.

Intelligence requirements and collection requirements are different objects. An intelligence requirement is what the consumer needs to know, phrased as a question about the world and tied to a decision. A collection requirement is what a collector must go get, phrased as a specific retrievable item from a named source. The decomposition chain runs decision, PIR, EEI, indicator, collection requirement, and each level answers a different question. Jumping from a PIR straight to searching produces browsing, not collection.

Planning is where legal and ethical review belongs. Reviewing after collection is a post-mortem. Every plan row passes the four tests, and the plan also carries a prohibited-activity statement, an individuals policy, an escalation path, and an operational security assessment. Planning is also where slow items get initiated, where the emphasis matrix decides where effort goes, and where the stopping condition is defined in advance.

Collection is disciplined execution, not searching. Log at the moment of collection, archive at the moment of collection, record negative results with their coverage boundaries, and follow the collection order: identity first, then geography, then structured official sources, then observational sources, then published sources, then user-generated material, with archives throughout. Running that order in reverse produces confidently wrong investigations, because unverified early material anchors everything read afterward. Stop when the requirement is satisfied, saturation is reached, the deadline arrives, or you are blocked, and say which one applied.

Processing makes material comparable and adds no meaning. Normalize time, names, units, and identifiers; deduplicate; resolve entities in both directions; translate, transcribe, and OCR; structure, tag, and index. Keep event dates and publication dates separate. Build the master timeline with a precision column. The test of complete processing is whether a second analyst could sort, group, and deduplicate your dataset without asking you anything.

Exploitation extracts what is not on the face of an item, and it is its own phase for a reason. Collapsed into processing it gets rushed; collapsed into analysis its derived findings get mixed with integrative judgments and become undefendable. Record every derived finding against its source item, with the method used and its own confidence, and separate observation from interpretation in every case. Chase every cross-reference, because in document-heavy work that is where most of the remaining value sits.

Analysis produces the first claim no single item makes. Assemble, generate at least three

hypotheses, test by disconfirmation, identify which items actually discriminate, surface assumptions, assign confidence with a stated reason, and rank gaps. In open source work, the defensible way to explain a confidence level is to state how many distinct categories support the finding and whether they could have been coordinated.

Dissemination is a set of decisions, not a document. Consumer, format, length, channel, timing, handling, retention, and confirmed receipt. Lead with the judgment, match the product to the consumer, characterize the record precisely, never assert intent, carry coverage boundaries into every negative, and apply the same standard to every party, because products are increasingly read by people with an interest in the outcome.

Feedback is the most skipped phase and the one that makes improvement possible. Utility feedback at one week asks whether anything changed; accuracy feedback after the forecast window asks whether the assessment was right. Score the reasoning as well as the outcome. Feedback loops back not only into the next product but into source ratings, assumption checks, formats, and the requirements process itself. When taking criticism, change the product when the critique concerns the consumer's needs, and hold the judgment when it concerns their preferences.

Continuous collection is architecturally different from repeated investigation. It requires a documented baseline, indicators defined in advance, a source cadence, written thresholds tied to defined actions, a triage and escalation path, and a review cycle that includes retiring requirements that no longer matter. Its three failure modes are alert fatigue, silent breakage, and baseline drift, and each has a specific structural remedy. An alert with no defined action should be deleted rather than tuned.

Everything traces. Decision to PIR to EEI to indicator to plan row to item to processed record to derived finding to assessment to product to feedback to revised requirement. Pick any sentence in a finished product and walk it backward; if a link is missing, the sentence is not defensible. That chain, plus the ethics annex recording what you declined to collect, is what separates professional intelligence work from confident research.

Part 5: Key Terms

Grouped by phase. Terms carried from earlier chapters are marked and restated briefly.

5.1 Methodology and Requirements

Intelligence collection methodology. The repeatable sequence of phases, each with defined inputs, outputs, and quality gates, that converts a decision maker's question into a delivered judgment and a running watch.

Quality gate. The pass condition for a phase; the work does not proceed until it is met.

Intelligence requirement. What the decision maker needs to know, phrased as a question about the world and tied to a decision.

Collection requirement. What a collector must go get, phrased as a specific retrievable item from a

named source.

Priority Intelligence Requirement (PIR) (Ch.1). A requirement the consumer has personally endorsed as a priority, which drives resource allocation.

Essential Element of Information (EEI) (Ch.1). A sub-question that must be answered for a PIR to resolve.

Indicator (Ch.1). An observable whose presence or absence helps answer an EEI. Defined in advance.

Decomposition chain. Decision, PIR, EEI, indicator, collection requirement. Each level answers a different question.

Answerability assessment. An explicit statement of which parts of a requirement open sources can and cannot reach, made before collection begins.

Requirement endorsement. The consumer's recorded approval of a requirement, without which downstream work has no owner.

Requirement revision history. The record of how a requirement changed during an effort, which prevents undocumented drift.

Consequential test. Whether a yes and a no would lead the consumer to different actions.

5.2 Planning

Collection plan. The document converting indicators into executable rows, each naming a category, a specific source, a query, a range, an access model, a legal note, an expected coverage boundary, a latency, an owner, and a priority.

Collection emphasis matrix. A cost-versus-likelihood grid used to decide where collection effort concentrates: quick wins, core effort, opportunistic, and last resort.

Slow item. A collection line with days or weeks of latency (records requests, imagery orders, interlibrary loans), which must be initiated at the start of an effort.

Four tests (Ch.3). Access, identity, protection, and purpose-and-harm; applied per plan row before any collection.

Prohibited-activity statement (Ch.2). A written enumeration, made in advance, of collection activities the engagement will not perform.

Individuals policy. The written rule governing which individuals may appear in an effort, in what capacity, and what will not be collected about them.

Escalation path. The named person who decides when a plan row's lawfulness or propriety is unclear.

Researcher OPSEC (Ch.2). Management of the collector's own detectable footprint, which protects the integrity of a lawful inquiry and never justifies concealment that enables improper collection.

Stopping condition. The written definition, set before collection, of what "done" looks like for each requirement.

5.3 Collection

Collection log. The per-item record: item ID, requirement ID, content summary, category, source, URL, query used, access date and time, archive location, publisher type and independence, reliability, credibility, coverage boundary, and collector.

Coverage boundary (Ch.3). The explicit statement of what a search did and did not cover, attached to every negative finding.

Negative result. A documented search returning nothing, recorded with its coverage boundary; a finding with real value.

Collection order. Identity, geography, structured official sources, observational sources, published sources, user-generated sources, with archives throughout.

Entity expansion (Ch.2). Enumerating all name variants, former names, subsidiaries, agents, officers, and jurisdictions before searching.

Jurisdiction determination (Ch.3). Establishing which county, municipality, and special districts govern a location, and therefore which records systems exist.

Collection drift. Following interesting threads that no requirement demanded.

Saturation. The point at which additional collection returns no new entity, date, or relationship for a given thread.

Manual collection. Person-executed query, reading, judgment, and recording.

Automated collection. Script, feed, alert, or scheduled retrieval, subject to terms of service, rate discipline, authorization, and retention of the script and parameters as part of the record.

Silent breakage. Automated collection that stops returning results without alerting, causing an absence of collection to be misread as an absence of activity.

Heartbeat check. A monitor that alerts on the absence of expected input.

5.4 Processing

Processing. Converting collected items into a comparable, indexed dataset. Adds no meaning.

Normalization (Ch.1). Rendering times, names, units, and identifiers into consistent formats.

Event date versus publication date. The date the described thing occurred versus the date the item was published; collapsing them destroys timelines.

Deduplication. Identifying items describing the same thing, marked rather than deleted.

Entity resolution (Ch.1). Determining that variant references are one entity, or that identical references are two, with a recorded basis for every merge.

Canonical entity list. The authoritative list of real-world entities with their observed variants.

Second identifier rule (Ch.3). A name match is never sufficient to merge; a registration number,

address, shared officer, or linking case is required.

Structuring. Extracting fielded data from unstructured items into a table.

Tagging. Applying a controlled vocabulary, including requirement and indicator IDs, to every item.

Master timeline. The sorted, sourced record of dated events, with a precision column and the count of supporting categories.

Precision (of a date). Whether a date is exact, day, month, quarter, or a bracketed window; carried forward rather than smoothed away.

De-identification. Removing or aggregating personal details, performed at processing rather than at dissemination.

5.5 Exploitation

Exploitation. Extracting from a single processed item information that is not on its face, recorded with the method used and its own confidence.

Derived finding. The output of an exploitation operation, recorded against its source item ID.

Observation versus interpretation. The discipline of writing what was seen separately from what it is judged to mean.

Geolocation (Ch.2). Determining where content was captured by matching fixed features.

Chronolocation (Ch.2). Determining when content was captured from shadows, weather, or dated features.

Metadata extraction (Ch.3). Reading a file's embedded data about itself; corroborating rather than conclusive.

Imagery interpretation. Deriving structures, activity, change, and dimensions from imagery, using a known reference for scale.

Document exploitation. Extracting entities, dates, amounts, obligations, and cross-references from a document.

Cross-reference rule. Every reference a document makes to another document becomes a new collection plan row.

Entity extraction. Pulling people, organizations, addresses, and identifiers from text into the canonical list.

Relationship extraction. Identifying connections between entities, with the stated basis recorded for each.

Technical enrichment. Deriving registration, resolution history, and related infrastructure from domains, certificates, and addresses.

5.6 Analysis

Analytic sequence. Assemble, generate hypotheses, test by disconfirmation, identify discriminators, surface assumptions, assign confidence, rank gaps, write.

Discriminating evidence. Items that distinguish between competing hypotheses, as opposed to items consistent with all of them.

Analysis of Competing Hypotheses (Ch.1). A technique evaluating all plausible explanations, scoring for disconfirmation rather than support.

Key Assumptions Check (Ch.1). Listing every assumption an assessment rests on and testing what happens if each is wrong.

Coverage assumption. The frequently unexamined premise that a record would exist if the event had occurred.

Confidence level (Ch.1). How solid the basis for a judgment is; distinct from probability.

Category-count heuristic. Explaining a confidence level by the number of distinct source categories supporting it and whether they could have been coordinated.

Estimative language (Ch.1). Standardized probability vocabulary from a published scale.

Intelligence gap (Ch.1). A specific unknown, stated with its analytic impact and the source that would resolve it.

Layering (Ch.1). Treating a prior product's low-confidence judgment as established background fact.

Politicization (Ch.1). Adjusting a judgment to match what the consumer wants to hear.

Narrative pull. The tendency to set aside untidy evidence because a clean story fits.

5.7 Dissemination, Feedback, and Continuous Collection

Dissemination decisions. Consumer, format, length, channel, timing, handling, retention, and confirmed receipt.

BLUF (Ch.1). Bottom line up front; the judgment in the first sentence.

Flash report. A one-paragraph product delivered immediately on an urgent finding.

Characterization discipline (Ch.3). Describing allegations, exhibits, testimony, orders, and settlements each in their correct evidentiary terms.

Ethics annex. The record of every individual encountered, what was deliberately not collected, how improper material was handled, and the retention decision.

Utility feedback. Consumer response days after delivery: was it read, did it answer the question, did anything change, what is the next question.

Accuracy feedback. Post-window scoring of whether a forecast resolved as assessed.

Accuracy record / forecast log. The logged, falsifiable forecast with its estimative term, confidence,

resolution date, and outcome.

Calibration. The measured relationship between an analyst's stated confidence and their realized accuracy.

Process scoring. Reviewing whether reasoning was sound given what was knowable, separately from whether the outcome matched.

Standing requirement. A continuing question with a review cadence rather than a deadline.

Baseline. The documented, dated description of normal against which change is detected.

Threshold. The pre-agreed level of observation that triggers a defined action.

Triage. The disposition of an incoming alert within a defined time by a named owner.

Escalation. Movement of a finding to a higher tier of consumer at a defined threshold.

Alert fatigue. Desensitization caused by low-value alerts, which causes the important alert to be ignored.

Baseline drift. Gradual change in normal conditions that is never re-baselined, causing monitoring to measure the wrong thing.

Scope creep (monitoring). Gradual collection of material beyond the standing requirement because it is easy.

Retirement (of a requirement). The deliberate, documented ending of a monitoring requirement that no longer serves a decision.

Documentation chain. The traceable path from decision to PIR to EEI to indicator to plan row to item to processed record to derived finding to assessment to product to feedback.

Audit test. Selecting any sentence in a product and walking it backward through the documentation chain.

Part 6: Knowledge Assessment

6.1 Short Answer (20 questions)

Answer in three to five sentences unless otherwise noted.

1. State the four reasons a methodology exists, and explain why the diagnosability reason is the one practitioners most underrate.
2. Name the nine phases in order and state the output of each in a few words.
3. Explain why exploitation is separated from processing and from analysis, and give one failure that results from each collapse.
4. Distinguish an intelligence requirement from a collection requirement, with an example of each

for the same investigation.

5. Write out the decomposition chain and explain what each level contributes.
6. State the four tests of a well-formed requirement and the failure each catches.
7. Explain the answerability assessment and why it should be delivered to the consumer early rather than late.
8. Describe the collection emphasis matrix and state which quadrant should absorb the bulk of the effort.
9. Explain why the legal and ethical review belongs in planning rather than in execution.
10. Explain what a slow item is and why it must be initiated on day one.
11. State the collection order and explain what goes wrong when it is run in reverse.
12. Explain the difference between the four stopping conditions, and name two things that are not stopping conditions.
13. Explain why processing adds no meaning, and state the test of whether processing is complete.
14. Explain the difference between an event date and a publication date, and what happens when they are collapsed.
15. Describe entity resolution in both directions and state the rule that governs merging.
16. Explain the observation-versus-interpretation discipline and why it matters when a finding is challenged.
17. State the cross-reference rule and explain why it is the highest-yield exploitation operation in records work.
18. Describe the analytic sequence and explain why identifying discriminating evidence is the step that separates trained analysts from careful readers.
19. Explain the two kinds of feedback, when each arrives, and what each improves.
20. Describe the three failure modes of continuous collection and the structural remedy for each.

6.2 Multiple Choice (15 questions)

1. The primary practical benefit of named lifecycle phases when an effort fails is that they: a) Assign blame accurately b) Allow the organization to locate which part broke c) Reduce the cost of collection d) Satisfy audit requirements
2. "Will Competitor X begin regional operations within 18 months?" is: a) A collection requirement b) An indicator c) An intelligence requirement d) An essential element of information
3. "Search the recorder's grantor/grantee index in four named counties for eight name variants over 24 months" is: a) An intelligence requirement b) A collection requirement c) A PIR d) An assumption
4. The legal and ethical review of a collection effort belongs in: a) Collection, as items are gathered b)

Planning, before anything is collected c) Analysis, when findings are weighed d) Dissemination, before publication

5. A slow item is best defined as: a) A low-priority collection line b) A collection line with days or weeks of latency, which must be initiated first c) A source that is difficult to read d) An item requiring translation

6. The correct collection order begins with: a) Social media, to establish current context b) News coverage, to establish the narrative c) Entity expansion and jurisdiction determination d) Satellite imagery, to establish physical facts

7. Processing is best described as the phase that: a) Extracts meaning from individual items b) Makes collected material comparable, adding no meaning c) Integrates findings into a judgment d) Delivers products to consumers

8. Which pair must be kept separate during processing? a) Reliability and credibility b) Event date and publication date c) Category and source d) Item ID and requirement ID

9. Geolocating a photograph and recording the matched features is an example of: a) Collection b) Processing c) Exploitation d) Dissemination

10. In the analytic sequence, identifying discriminating evidence means: a) Finding the evidence that best supports the leading hypothesis b) Finding the items that actually distinguish between competing hypotheses c) Removing items of low reliability d) Counting the number of supporting sources

11. In OSINT, the most defensible way to explain a high confidence level is: a) The number of items collected b) The number of distinct source categories supporting it, and whether they could have been coordinated c) The reputation of the primary source d) The length of the investigation

12. Utility feedback asks primarily: a) Whether the forecast proved correct b) Whether anything changed as a result of the product c) Whether the sourcing was adequate d) Whether the analyst followed the methodology

13. An alert type that has no defined action should be: a) Tuned to fire less often b) Routed to a different consumer c) Deleted d) Aggregated into a weekly summary

14. Silent breakage in continuous collection refers to: a) An analyst failing to report a finding b) A collection source failing without alerting, so absence of collection is misread as absence of activity c) A consumer ignoring an alert d) A threshold set too high

15. A consumer asks you to soften a judgment because it will be unwelcome to the board. The correct response is: a) Revise the judgment, since the consumer owns the requirement b) Revise the format and language, but hold the judgment c) Withdraw the product d) Add supporting evidence you did not previously use

6.3 True or False (10 questions)

Mark true or false. For each false statement, write one sentence explaining what makes it false.

1. Under time pressure, the correct compression is to delete phases that do not fit the schedule.

2. A collection requirement should be executable by someone who knows nothing about the case.
 3. Reviewing the legality of collection after it has occurred satisfies the planning gate.
 4. A search returning no results is not a finding and need not be recorded.
 5. Processing adds interpretation to collected material.
 6. A name match alone is sufficient basis for merging two entity references during processing.
 7. Every cross-reference encountered during document exploitation should become a collection plan row or be explicitly closed with a reason.
 8. Most collected evidence in a typical investigation discriminates between competing hypotheses.
 9. Accuracy feedback and utility feedback arrive at different times and improve different things.
 10. Retiring a monitoring requirement that no longer supports a decision is a legitimate and valuable act.
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Part 7: Critical Thinking Exercises

Each group targets one methodology skill. Where an exercise asks for a written artifact, produce it in the format the chapter specifies.

Defining intelligence requirements

7.1 A hospital system's chief operating officer says: "I want to know if we're at risk from the new urgent care chain." Apply the four tests, identify which fail, and rewrite as a PIR with three EEIs. Then state which part of the original concern open sources cannot reach, and how you reframed around it.

7.2 A county emergency manager asks for "situational awareness on the river." Write two different PIRs that this could mean, tied to two different decisions, and explain why the choice between them must be made by the consumer rather than by you.

7.3 You are given five requirements and can complete two. Design the prioritization you would present, using the two-dimensional grid from Section 1.2.4, and write the sentence you would use to tell the consumer that their highest-priority question cannot be answered from open sources.

Developing collection plans

7.4 For the PIR you wrote in 7.1, produce four collection plan rows in full, including the legal note and the expected coverage boundary for each. Then identify which of the four is a slow item and what happens if it is initiated in week three instead of week one.

7.5 A plan row reads: "Search social media for employee comments about the competitor's expansion." Assess this row against the four tests and the individuals policy. Rewrite it into something defensible, or state why it should be removed.

7.6 Your plan has 22 rows and you have 12 hours. Apply the emphasis matrix and state which rows

you execute, which you defer, and which you drop, with the reasoning for each category.

Evaluating source quality

7.7 Two items support the same finding: a state regulatory filing and a trade press article that cites "company officials." Rate each for reliability and credibility, explain the difference in terms of production process, and write the sentence you would use in a product.

7.8 Six items support a finding, all from the same category. Explain why this is weaker than three items from three categories, and describe how you would word the confidence statement.

7.9 A source that has been reliable across four prior investigations produces a claim that conflicts with two compelled records. Explain how you handle this, and what you record about the source afterward.

Organizing and processing collected data

7.10 You have 140 items with dates recorded in four different formats, three time zones, and two fiscal calendars. Describe your processing sequence and state, specifically, what analytic error each normalization step prevents.

7.11 Your dataset contains "Ridgeline Env. Svcs," "Ridgeline Environmental Services LLC," "Ridgeline Environmental Services of the Northwest LLC," and "Kiln Road Transport LLC." Describe how you resolve these, what second identifier you would require for each merge, and which you would leave flagged as unresolved.

7.12 A colleague proposes deleting duplicate items to keep the dataset clean. Explain why duplicates are marked rather than deleted, and what is lost if they are removed.

Analyzing evidence objectively

7.13 You have a leading hypothesis supported by 31 of your 40 items. Explain why that ratio is close to meaningless, and describe how you would identify which items actually carry the judgment.

7.14 Your assessment rests on the assumption that a permit would exist if the construction had occurred. Name this assumption class, describe how you would test it, and state what your confidence should be if you cannot.

7.15 Halfway through analysis you notice that you have been reading every ambiguous item as supporting your initial theory. Name the bias, describe the specific control you would apply now, and explain what you would tell the consumer if the control changes your judgment.

Producing intelligence conclusions

7.16 Write a four-sentence bottom line for a finding of your choice that contains a judgment, a confidence level with its reason, estimative language, and a distinction between what requires action now and what informs a later decision.

7.17 Your finding is that two entities share a manager and an address, contrary to a representation of independence. Write it three ways: as a fact, as an assessment, and as an improper allegation. Explain

what makes the third improper and what legal exposure it creates.

7.18 A consumer with six minutes needs the same content that a legal team needs in full. Describe the two products you would produce, and state what is in one and not the other.

Recommending improvements through feedback

7.19 Your product was delivered on time, was accurate, and changed nothing. Work through the four diagnoses this could indicate, describe the feedback questions that would distinguish them, and give the remedy for each.

7.20 A forecast you made at high confidence did not occur. Describe how you would score it on both the outcome and the process dimensions, and state what system change you would recommend if the reasoning was sound but the source was wrong.

7.21 Design a continuous monitoring plan for a standing requirement of your choice: baseline, four indicators, sources and cadence, three thresholds with defined actions, and a review cycle. Then state the condition under which you would recommend retiring it.

Part 8: Hands-On Lab Exercises

General Lab Guidance

These five labs are one continuous investigation. Lab 1's output is Lab 2's input, and so on. Do them in order. Keep every artifact; Lab 5 requires all of them.

Rules 1 through 5 apply throughout. No private individuals. Public means genuinely public. Document as you go. No dark web. Automation only where permitted, disclosed, and rate-limited.

Rubric scale. 0 to 4 per criterion:

Score	Meaning
4	Professional quality; deliverable to a real consumer unchanged
3	Competent; minor weaknesses
2	Developing; partial compliance, notable gaps
1	Attempted; concept recognized but misapplied
0	Missing or fundamentally incorrect

Universal requirements, graded in every lab. Every judgment carries a confidence level with a stated reason. Every product names at least one ranked gap. Every finding appears in the collection log with its coverage boundary. Every artifact carries its requirement ID.

The Chained Scenario

Your role. You are the sole open source analyst at **Kestrel Ridge Partners**, a fictional twelve-person advisory firm. Kestrel Ridge is fictional. Its clients are fictional. **The organization you will investigate is real, and it must be an organization, never an individual.**

Your consumer. **Dana Ruiz**, Kestrel Ridge's fictional managing director, who will act as the stakeholder in Lab 5. If you are working in a class, a classmate or instructor plays Dana. If you are working alone, you will brief a willing listener and record their questions, or complete the structured self-critique variant described in Lab 5.

Subject selection rules. Choose one real organization that is:

- Institutional: a publicly traded company, a public agency, a university, a municipality, a utility, a port authority, a transit agency, a hospital system, or a large nonprofit.
- Documented across at least six of the Chapter 3 categories.
- Not a private individual, not a sole proprietorship, and not an organization where the coverage would center on one identifiable person.
- Not your own employer, and not an organization where you have non-public knowledge. This matters: you must be able to state honestly that everything in your product came from public sources.

Choose your client decision from this menu. Each is a template producing an answerable requirement. Fill in your subject.

#	Fictional client decision
A	Our client is considering a multi-year supply contract with [subject]. Should they proceed, renegotiate terms, or decline?
B	Our client is considering leasing space adjacent to [subject]'s facility. What is the operational and regulatory picture at that site?
C	Our client is a municipality considering a partnership with [subject]. What does the public record show about performance and obligations?
D	Our client competes with [subject] and must decide whether to expand into [region]. Is [subject] expanding there?
E	Our client insures facilities like [subject]'s. What does the public record show about the risk profile of its principal site?
F	Our client is a public-interest funder deciding whether to grant to [subject]. What does the public record show about capacity and governance?

A note on tone. You are not building a case against anyone. Several of these decisions could produce a favorable finding, and a well-run investigation frequently does. The professional standard is the same either way.

Lab 1: Intelligence Requirements and Collection Objectives

Time required: 2 to 2.5 hours **Format:** Individual **Difficulty:** Foundational to intermediate

Objectives

1. Convert a vague client concern into an endorsed, answerable requirement.
2. Decompose a PIR into EEIs and observable indicators.
3. Produce an honest answerability assessment.
4. Prioritize requirements against a resource constraint.

Prerequisites

- Sections 1.1 and 1.2.
- Subject and client decision selected per the scenario rules.
- No collection yet. This lab is planning only, and beginning to search is a rubric deduction.

Steps

Step 1 (15 min). Write the decision statement: the decision, the decision maker (Dana Ruiz, on behalf of the fictional client), the resources or commitment at stake, and the deadline. Set the deadline four weeks out. Then write one sentence describing what the client would do differently under a yes versus a no.

Step 2 (30 min). Write **three PIRs**. Apply the four tests to each on Worksheet 1A and record which tests each passes. Revise any that fail until all three pass, or drop and replace them. Record the failed drafts; the revision history is graded.

Step 3 (40 min). Decompose each PIR into three to five EEIs, and each EEI into two to four indicators, on Worksheet 1B. Indicators must be observable: something that could actually appear in a source. "Their internal strategy" is not an indicator. "A recorded lease memorandum naming the entity" is.

Step 4 (20 min). Write the answerability assessment on Worksheet 1C. For each PIR: which parts open sources can reach, which they cannot, and for the unreachable parts, what non-open source or process would answer it. At least one genuine limitation must appear. If you believe all three PIRs are fully answerable from open sources, you have almost certainly written a requirement about observable behavior and are treating it as though it answered a question about intent.

Step 5 (20 min). Prioritize on Worksheet 1D using the two-dimensional grid. Rank the three PIRs,

state the reason, and write the two sentences you would send Dana explaining the ranking and flagging anything that cannot be answered from open sources.

Step 6 (15 min). Write the requirement record (Section 1.2.5 fields) for your top-priority PIR. This is the document that every later lab traces back to.

Step 7 (10 min). Self-check against the requirements quality gate: does each PIR name a decision, is each answerable, is each deadline-bound, and would each change what the consumer does? Record any that you are unsure about and why.

Worksheet 1A: Requirement Tests

PIR draft	Decision-linked?	Answerable?	Deadline-bound?	Consequential?	Verdict	Revision
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Worksheet 1B: Decomposition

PIR #	EEI #	EEI statement	Indicator #	Indicator (observable)	Likely category
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Worksheet 1C: Answerability

PIR #	Reachable from open sources	Not reachable	What would answer the unreachable part
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Worksheet 1D: Prioritization

PIR #	Decision impact (H/L)	Open-source answerable (Y/N)	Quadrant	Rank	Reason
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Expected Outcomes

- Three PIRs, all passing four tests, with at least one visible revision from a failed draft. Students who submit three perfect first drafts have usually written vague requirements and marked them passing.
- Twelve or more indicators total, the majority genuinely observable. "Their plans," "their intentions," and "their internal discussions" are the common non-observable failures.
- At least one honest unanswerable element, with the non-open source named.
- A prioritization that reflects a resource constraint rather than listing all three as high.
- A requirement record complete enough that a stranger could pick up the investigation from it.

Discussion Questions

1. Which PIR draft failed a test, and what did the failure reveal about what you actually wanted to know?
2. You reframed an intent question into an observable-precursor question. What did you lose in that reframing, and is the loss acceptable?
3. Your top-priority PIR is ranked first. What would have to change for the ranking to flip?
4. Write the sentence you would send Dana if all three PIRs were unanswerable from open sources.

Grading Rubric: Lab 1 (44 points)

Criterion	Weight	4 points looks like
Decision statement precision	x2	Decision, decision maker, stake, deadline, and the yes-versus-no difference
PIR quality against four tests	x3	All three pass, tests applied honestly
Visible revision history	x1	Failed drafts retained with the reason
EEI decomposition	x2	Collectively sufficient, non-overlapping, traceable
Indicator observability	x3	Twelve-plus, genuinely observable, category noted
Answerability assessment	x2	Honest, with the non-open source named for each gap
Prioritization	x2	Grid applied; constraint respected; consumer sentence written
Requirement record completeness	x1	All fields; usable by a stranger

Total possible: 64 weighted. Passing: 44.

Lab 2: The Complete Collection Plan

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Intermediate

Objectives

1. Convert indicators into executable collection plan rows.

2. Apply legal and ethical tests per row, before collection.
3. Anticipate coverage boundaries and latency.
4. Sequence work using the emphasis matrix and identify slow items.
5. Write a prohibited-activity statement and an individuals policy.

Prerequisites

- Sections 1.3 and Chapter 3 in full.
- Lab 1 complete. Bring your top-priority PIR and its indicators.
- Still no collection. Confirming that a source system exists is permitted; querying it is not.

Steps

Step 1 (20 min). Entity expansion and jurisdiction determination on Worksheet 2A. Before any source planning, enumerate your subject's legal name, former names, trade names, subsidiaries and affiliates, the jurisdictions where it operates or is registered, and for its principal site the county, municipality, and any special districts. You may confirm these from a registry or map without treating it as collection; log what you did.

Step 2 (75 min). Build the collection plan on Worksheet 2B. One row per indicator, minimum fourteen rows, using every column from Section 1.3.1. Requirements for full credit:

- At least six of the Chapter 3 categories represented.
- At least four rows from the official and compelled group (government records, court records, corporate filings, financial records, public databases).
- At least one archived-website row.
- At least two slow items identified as such.
- Every row's query written specifically enough that a stranger could execute it.
- Every row's expected coverage boundary written **before** collection.

Step 3 (25 min). Legal and ethical review on Worksheet 2C. Apply the four tests to every row. At least one row in a realistic plan should fail or require narrowing; if none of yours do, examine your social media and individual-related rows again. Record the disposition of every row that failed.

Step 4 (15 min). Write the prohibited-activity statement (minimum five specific items) and the individuals policy (who may appear, in what capacity, and what will not be collected).

Step 5 (20 min). Write the OPSEC assessment on Worksheet 2D: which rows are visible to the subject, what you will do about it, and where your research will and will not be conducted from.

Step 6 (20 min). Sequencing on Worksheet 2E. Place every row in the emphasis matrix. Then produce an execution order for Lab 3's four hours, naming which slow items are initiated first, which rows you will execute, and which you will defer. Write your stopping condition for each EEI.

Step 7 (10 min). Self-check against the planning quality gate.

Worksheet 2A: Entity and Jurisdiction

Legal name	Former names	Trade names	Affiliates	Jurisdictions	County	Municipality	Special districts
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Worksheet 2B: Collection Plan

Ind. ID	Category	Specific source/system	Query or approach	Range	Access and cost	Legal note	Expected coverage boundary	Latency	Owner	Priority	Status
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Worksheet 2C: Four Tests Per Row

Row	Access	Identity	Protection	Purpose/harm	Verdict	Disposition if failed
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Worksheet 2D: OPSEC

Row	Visible to subject?	How	Mitigation	Decision
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Worksheet 2E: Emphasis and Sequence

Row	Cost (L/H)	Likelihood source holds it (L/H)	Quadrant	Execution order	Slow item?
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Expected Outcomes

- Fourteen or more rows across six or more categories, four or more from official and compelled sources.
- Queries specific enough to hand to a stranger. "Search public records" scores 1; "county recorder grantor/grantee index, search these six name variants, 24-month range" scores 4.
- Coverage boundaries written in advance. This is the criterion students most often skip and it carries heavy weight.
- Two or more slow items identified and scheduled first.
- At least one row narrowed or removed by the legal and ethical review, with the reasoning

recorded.

- A prohibited-activity statement that is specific to this engagement rather than a generic restatement of the rules.

Discussion Questions

1. Which row did the four tests change, and what would have happened if you had discovered the problem during collection instead?
2. Your slow items have a latency of days or weeks. What does your plan look like if the deadline were one week rather than four?
3. Which quadrant holds most of your rows? Is that the right distribution for this requirement?
4. You wrote coverage boundaries before collecting. Which one do you expect to turn out wrong, and why?

Grading Rubric: Lab 2 (68 points)

Criterion	Weight	4 points looks like
Entity expansion and jurisdiction	x3	Complete; former names, affiliates, and all governing jurisdictions identified
Plan breadth	x2	Fourteen-plus rows, six-plus categories, four-plus official and compelled
Query specificity	x3	Every row executable by a stranger
Coverage boundaries written in advance	x3	Present and realistic for every row
Latency and cost realism	x2	Accurate; slow items correctly identified
Four tests applied per row	x3	Every row assessed; at least one narrowed or removed with reasoning
Prohibited-activity statement	x2	Five-plus items, specific to this engagement
Individuals policy	x2	Clear on capacity, exclusions, and minors
OPSEC assessment	x1	Visibility identified with realistic mitigation
Emphasis matrix and sequencing	x2	All rows placed; execution order defensible; slow items first
Stopping conditions	x1	Written per EEI, in advance

Total possible: 96 weighted. Passing: 66.

Lab 3: Collection and Processing

Time required: 3.5 to 4.5 hours **Format:** Individual **Difficulty:** Intermediate

Objectives

1. Execute a collection plan with full documentation discipline.
2. Record negative results with coverage boundaries.
3. Normalize, deduplicate, resolve entities, and index a mixed dataset.
4. Build a master timeline with a precision column.
5. Recognize and record saturation, deviation, and blocking.

Prerequisites

- Sections 1.4 and 1.5.
- Labs 1 and 2 complete. Bring the plan.
- Internet access. A spreadsheet application.

Steps

Step 1 (10 min). Initiate slow items. If a records request, document order, or interlibrary request is in your plan, send it now and log the send. You will likely not receive it within this lab, and that is the point: record it as a gap with an expected arrival date.

Step 2 (140 min). Execute the plan in the collection order from Section 1.4.3: identity, geography, structured official sources, observational sources, published sources, user-generated sources, archives throughout. Log every item as you collect it on Worksheet 3A, to the full standard. Requirements:

- Minimum 30 logged items.
- Minimum 6 categories represented.
- Minimum 4 negative results, each with a coverage boundary.
- Every item archived at collection, with the archive location recorded.
- Every deviation from the plan recorded with the reason.

Step 3 (15 min). Deviation and stopping record on Worksheet 3B. For each EEI: which stopping condition applied (satisfied, saturated, deadline, blocked), the evidence for it, and what remains open.

Step 4 (60 min). Process the dataset into Worksheet 3C, using the minimum viable schema from Section 1.5.3. Perform, and record:

- Time normalization, with time zones recorded.
- Separation of event date from publication date for every item.
- Name and identifier normalization.
- Deduplication: mark duplicates, do not delete.
- Entity resolution: build the canonical entity list on Worksheet 3D with every observed variant and the basis for each merge.
- Tagging: every item to at least one indicator.

Step 5 (30 min). Build the master timeline on Worksheet 3E, with the precision column and the count of distinct supporting categories per row.

Step 6 (15 min). Processing quality gate self-check against the Section 1.5.6 list. Record any gate you did not pass and why.

Step 7 (10 min). Write the collection summary: how many items, from which categories, which EEIs are covered and which are not, what the slow items are, and what the largest coverage gap is at this point.

Worksheet 3A: Collection Log

Item ID	Ind. ID	Content summary	Category	Source/system	URL	Query used	Access date/time (TZ)	Archive location	Publisher type	Independent?	Rel.	Cred.	Coverage boundary	Collector
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Worksheet 3B: Deviation and Stopping

EEI	Planned rows executed	Deviations and reason	Stopping condition applied	Evidence for it	What remains open
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Worksheet 3C: Processed Dataset

Item ID	Ind. ID	Type	Category	Source	Event date	Pub. date	Access date/time	Entities (canonical)	Location	Duplicate of	Independence	Rel./Cred.	Archive	Processing notes
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Worksheet 3D: Canonical Entity List

Canonical name	Observed variants	Type	Basis for merge	Unresolved candidates
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Worksheet 3E: Master Timeline

Date	Precision	Event (as observed)	Entities	Supporting item IDs	Distinct categories	Confidence in event	Notes
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Expected Outcomes

- Thirty-plus items across six-plus categories, all logged at the moment of collection. Retroactive logging is usually visible in the timestamps and is heavily penalized.
- Four-plus negative results with coverage boundaries. Students frequently record zero, which is the most common failure in this lab.
- At least one deviation from the plan, honestly recorded. Plans always change; a submission showing perfect adherence is usually a submission where the log was written afterward.
- Duplicates marked rather than deleted, including at least one case where two items describe the same event.
- A canonical entity list with at least three variants resolved and a recorded basis for each merge, plus at least one candidate left flagged as unresolved.
- A master timeline where the precision column is genuinely used: a mix of exact dates, month-level entries, and at least one bracketed window.
- A collection summary that names the largest coverage gap rather than presenting the collection as complete.

Discussion Questions

1. Which negative result was most informative, and how would you word it in a product so it cannot be misread as a conclusion?
2. Where did you deviate from the plan? Was the deviation justified, and how would you have planned differently?
3. Which entity resolution decision were you least sure about? What second identifier would settle it?
4. Your timeline has entries at different precisions. Which imprecise entry matters most, and what would sharpen it?
5. Did you reach saturation on any EEI? What evidence told you?

Grading Rubric: Lab 3 (76 points)

Criterion	Weight	4 points looks like
Collection volume and breadth	x2	Thirty-plus items, six-plus categories
Collection order followed	x2	Identity and geography first; user-generated last
Log completeness	x3	Every field, every item, contemporaneous
Archiving at collection	x3	Every item archived, location recorded and retrievable
Negative results with boundaries	x3	Four-plus, precisely bounded
Deviation and stopping record	x2	Honest, with the stopping condition named per EEI
Time normalization and dual dates	x3	Event and publication dates separated throughout, zones recorded
Deduplication	x2	Marked, not deleted; at least one real duplicate found
Entity resolution	x3	Canonical list with variants, merge basis recorded, unresolved flagged
Tagging to indicators	x2	Every item traced to an indicator
Master timeline with precision	x3	Genuine precision variation; supporting categories counted
Collection summary honesty	x2	Names the largest gap rather than claiming completeness

Total possible: 120 weighted. Passing: 82.

Lab 4: Exploitation, Analysis, and Assessment

Time required: 3 to 3.5 hours **Format:** Individual **Difficulty:** Advanced

Objectives

1. Exploit processed items to produce derived findings with stated methods.
2. Separate observation from interpretation in every derivation.
3. Test competing hypotheses by disconfirmation and identify discriminating evidence.
4. Assign calibrated confidence with a stated reason.

5. Identify and rank intelligence gaps.

Prerequisites

- Sections 1.6 and 1.7.
- Lab 3 complete. Bring the processed dataset, canonical entity list, and master timeline.

Steps

Step 1 (50 min). Exploitation. Select at least eight items and perform at least four different exploitation operations across them (Section 1.6.2). Record each derived finding on Worksheet 4A with its source item ID, the method used, the observation, the interpretation, and the confidence in the derivation.

Requirements:

- At least one geolocation, imagery interpretation, or metadata extraction.
- At least one document exploitation producing cross-references.
- At least one relationship extraction with the stated basis for each edge.
- Every derived finding written in the two-column observation and interpretation form.

Step 2 (15 min). Cross-reference harvest on Worksheet 4B. List every cross-reference encountered during exploitation. For each: add it to the collection plan as a new row, or close it with a reason. At least three cross-references should appear; if none did, you exploited only summary-level material and should return to Step 1 with a document-heavy item.

Step 3 (20 min). Build the relationship map on Worksheet 4C: entities as nodes, relationships as edges, each edge carrying its supporting item IDs and its basis. Draw it in text or as a diagram. Mark any edge whose basis is a single item in a single category.

Step 4 (40 min). Hypothesis testing on Worksheet 4D. Write at least three hypotheses that answer your top-priority PIR, including the one you consider least likely, stated fairly. Score every relevant item and derived finding against each hypothesis as supporting, contradicting, or neutral. **Score for disconfirmation:** for each hypothesis, ask what evidence would disprove it and whether you looked for that evidence.

Step 5 (15 min). Identify discriminating evidence on Worksheet 4E: which items or findings actually distinguish between hypotheses. Then state, in one sentence, how many items your judgment actually rests on. Most students discover the number is small, which is the intended lesson.

Step 6 (20 min). Key assumptions check on Worksheet 4F. List at least five assumptions, including at least one coverage assumption ("a record would exist if this had happened"). For each: how you would test it, whether you did, and what changes if it is wrong.

Step 7 (25 min). Write the assessment on Worksheet 4G: the judgment, the confidence level with the category-count reasoning, the alternatives rejected and why, the discriminating evidence, the assumptions, and the gaps ranked by analytic impact with the resolving source named for each.

Step 8 (10 min). Analysis quality gate self-check against the Section 1.7.6 list.

Worksheet 4A: Derived Findings

Finding ID	Source item ID	Operation	Method detail	Observation	Interpretation	Confidence in derivation
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Worksheet 4B: Cross-Reference Harvest

Reference found	In which item	Document sought	Added to plan or closed	Reason if closed
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Worksheet 4C: Relationship Map

From	To	Relationship	Basis	Supporting item IDs	Categories	Single-source edge?
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Worksheet 4D: Competing Hypotheses

Item / finding	H1	H2	H3	Notes
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For each hypothesis: what would disprove it, and did I look?

Worksheet 4E: Discriminating Evidence

Item / finding	Which hypotheses does it distinguish?	Why
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Worksheet 4F: Key Assumptions

Assumption	Type	How I would test it	Did I test it?	What changes if wrong
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Worksheet 4G: Assessment

Use the Section 1.8.3 structure, one page maximum.

Expected Outcomes

- Eight-plus items exploited across four-plus operations, every derived finding in two-column form with its method.
- Three-plus cross-references harvested and dispositioned.
- A relationship map where at least one edge is flagged as single-source.

- Three genuine hypotheses, including a fairly stated least-likely one. Straw-man alternatives are the most common failure in this lab and score 1.
- A disconfirmation question answered honestly for each hypothesis, including at least one "I did not look for that" admission where true.
- Discriminating evidence identified, with the count of items the judgment rests on stated explicitly. A judgment resting on two or three items is normal and honest; a student who claims all thirty items are discriminating has not understood the step.
- Five-plus assumptions including a coverage assumption.
- Confidence that is defensible on the category-count reasoning, and in most cases lower than the student's initial instinct.
- Gaps ranked by analytic impact, each with a named resolving source.

Discussion Questions

1. How many items does your judgment actually rest on? How does that compare with how many you collected?
2. Which hypothesis did you find hardest to state fairly, and what does that tell you?
3. Name an assumption you did not test. What confidence penalty should that carry?
4. Which derived finding would collapse your assessment if the method were wrong? How would you verify it?
5. Your confidence probably moved during this lab. In which direction, and why is that direction usually the right one?

Grading Rubric: Lab 4 (88 points)

Criterion	Weight	4 points looks like
Exploitation breadth	x2	Eight-plus items, four-plus operations
Method recorded per derivation	x3	Specific enough to be repeated by another person
Observation versus interpretation	x3	Two-column discipline maintained without exception
Cross-reference harvest	x2	Three-plus, each dispositioned
Relationship map with edge basis	x2	Every edge sourced; single-source edges flagged
Hypothesis quality	x3	Three genuine hypotheses, least-likely stated fairly
Disconfirmation scoring	x3	What-would-disprove answered per hypothesis, honestly

Criterion	Weight	4 points looks like
Discriminating evidence identified	x3	Correctly isolated; count stated explicitly
Key assumptions	x2	Five-plus including a coverage assumption, with consequences
Confidence calibration	x3	Category-count reasoning; overconfidence penalized
Ranked gaps with resolving sources	x2	Ranked by analytic impact, each with a named source
Assessment structure	x2	Section 1.8.3 structure, one page, facts separated

Total possible: 120 weighted. Passing: 82.

Lab 5: Production, Briefing, Feedback, and Revision

Time required: 3 to 3.5 hours plus the briefing **Format:** Individual, with a partner, instructor, or listener as the stakeholder **Difficulty:** Advanced

Objectives

1. Produce a professional intelligence product matched to a named consumer.
2. Brief the product and answer questions without overstating.
3. Collect structured feedback.
4. Revise appropriately, distinguishing critiques of the product from critiques of the judgment.
5. Design continuous monitoring for the standing part of the requirement.

Prerequisites

- Sections 1.8, 1.9, and 1.10.
- Labs 1 through 4 complete.
- A person to play Dana Ruiz for 20 minutes. If none is available, use the self-critique variant in Step 4b.

Steps

Step 1 (50 min). Write **Product A**, the primary deliverable: a two-page assessment for Dana Ruiz, using the Section 1.8.3 structure, with the collection log and ethics annex as annexes that do not count toward the two pages. Every fact carries its category, source, and ratings. Every judgment carries a confidence level with its reason.

Step 2 (25 min). Write **Product B**, a one-paragraph flash-format version for a hypothetical consumer with ninety seconds: the same judgment, the same confidence, the action implication, and nothing else. Then write two sentences on what you removed and why removing it was defensible.

Step 3 (20 min). Prepare and deliver the briefing, five minutes maximum, following Section 1.8.4: judgment first, two or three discriminating facts, the rejected alternative and why, gaps and what would close them, then stop. Do not read the product aloud.

Step 4a (20 min). Question period and feedback. The stakeholder asks questions for ten minutes and then completes Worksheet 5A, the four utility questions plus a request for two specific changes. Record the questions asked verbatim; the questions are data about what the requirement actually was.

Step 4b (alternative, 25 min). If no stakeholder is available: leave the product untouched for at least four hours, then complete Worksheet 5A yourself against a written checklist rather than by rereading impressionistically, and additionally write the three hardest questions a skeptical consumer would ask and answer them in writing. Note in your submission that you used the self-critique variant; it is a weaker control and should be named as one.

Step 5 (30 min). Revise. Complete Worksheet 5B: for each piece of feedback, classify it as a product critique (format, length, clarity, framing) or a judgment critique (the assessment itself), state your response, and state what changed. **At least one item of feedback should be accepted and at least one should be respectfully declined**, with the reasoning recorded. Then produce Product A, revision 2.

Step 6 (25 min). Continuous monitoring design on Worksheet 5C: the standing requirement, the baseline as of today, four indicators with their properties tested (observable, attributable, discriminating, timely, sustainable), sources and cadence, three thresholds with defined actions and owners, the review cycle, and the retirement condition.

Step 7 (15 min). Accuracy record on Worksheet 5D: state your central forecast falsifiably, with its estimative term, confidence, and resolution date. Then write, in two sentences, how you would score yourself when the resolution date arrives.

Step 8 (10 min). Methodology retrospective, half a page: which phase was weakest in your investigation, what specifically you would do differently, and one change to your own process you will carry into the capstone.

Worksheet 5A: Stakeholder Feedback

Question	Response
Did you read it? If not, why not?	
Did it answer the question you actually had?	
Did anything change as a result?	
What question do you have now?	
Two specific changes requested	
Questions asked during the briefing (verbatim)	

Worksheet 5B: Feedback Disposition

Feedback item	Product critique or judgment critique?	Response	What changed	Reasoning if declined
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Worksheet 5C: Continuous Monitoring Plan

Element	Content
Standing requirement	
Baseline (dated)	
Indicator 1 to 4, with property tests	
Source and cadence per indicator	
Threshold tiers, actions, owners	
Review cycle	
Retirement condition	

Worksheet 5D: Accuracy Record

Field	Content
Forecast, stated falsifiably	
Estimative term and probability band	
Confidence and its reason	
Resolution date	
How I will score it	

Expected Outcomes

- Product A within two pages, with annexes, judgment in the first sentence, facts separated from assessment, every fact rated and categorized.
- Product B genuinely shorter, not a compressed summary of everything, and a defensible account of what was removed.
- A briefing that opens with the judgment. Students who open with methodology or background almost always run out of time before reaching the judgment, which is the lesson.
- Stakeholder questions recorded verbatim, and at least one question revealing that the consumer's real interest differed from the stated requirement. This is the most common and most valuable outcome of the briefing.

- Feedback correctly classified, with at least one acceptance and at least one respectful decline. A student who accepts every critique including a request to soften the judgment has failed the central lesson of this lab.
- A monitoring plan where every indicator passes the five property tests and every threshold has a defined action and owner.
- A falsifiable forecast with a resolution date.
- A retrospective naming a real weakness rather than a generic one.

Discussion Questions

1. What did the stakeholder's questions reveal about the requirement you were actually answering?
2. Which feedback did you decline, and how did you word the decline?
3. Product B removed most of Product A. What did the removal cost, and who would that cost matter to?
4. Your monitoring plan has four indicators. Which is most likely to produce false positives, and what threshold protects against that?
5. In your retrospective, which phase was weakest? Was it weak because of skill, time, or planning?

Grading Rubric: Lab 5 (84 points)

Criterion	Weight	4 points looks like
Product A structure and BLUF	x3	Judgment and confidence in the first sentence; full structure
Fact and assessment separation, with ratings	x3	Consistent throughout; every fact categorized and rated
Length discipline and annexes	x2	Two pages plus annexes; nothing important pushed out
Product B compression	x2	Genuinely brief; removal defended
Briefing structure	x3	Judgment first, discriminators, rejected alternative, gaps, then stops
Question handling	x2	Distinguishes observation from assessment; says "I do not know" where true
Feedback captured	x2	All four questions plus verbatim briefing questions
Feedback classification	x3	Product versus judgment critique correctly distinguished

Criterion	Weight	4 points looks like
Revision quality	x2	Accepted changes implemented; declined changes reasoned
At least one respectful decline	x2	Present, well worded, and correctly grounded
Monitoring plan completeness	x3	Baseline, indicators tested against five properties, thresholds with actions and owners, review cycle, retirement condition
Falsifiable forecast	x2	Stated with estimative term, confidence, and resolution date
Retrospective honesty	x2	Specific weakness identified with a concrete change

Total possible: 124 weighted. Passing: 86.

Part 9: Capstone Assignment

Operation Clearline: A Statewide Solar Contractor Investigation

Weight: 35% of chapter grade **Time required:** 6 to 9 hours **Format:** Individual, across multiple sittings **Deliverables:** Nine items, one per phase, listed in Section 9.4

9.1 The Scenario

You are the open source analyst for the **Consumer Protection Division of a state Attorney General's office**. Everything in this scenario is fictional.

The situation. Over eight months the Division has received 212 complaints concerning residential solar installation. The complaints cluster around four themes: financing terms that differ from what was represented at the door, systems that were never energized, warranty and service contacts that go unanswered, and liens appearing on homes that the homeowners say they did not knowingly grant.

The Division Chief has 3 investigators and 1 attorney available for this. She has to choose between three courses of action within six weeks:

1. Continue handling complaints individually.
2. Build a pattern investigation against one or more connected operations.
3. Refer the matter to the state licensing board and the financial regulator, and focus Division resources elsewhere.

Complications.

- Complaint volume is still rising, so a slow answer costs real harm.
- A trade association has publicly stated that the complaints reflect a small number of bad actors and that broad action would harm legitimate installers, which is a claim you must assess on the same standard as any other.
- Two of the named businesses have already ceased operating.
- Every complainant is a private individual, and the Division is subject to public records law, meaning your product may become public.

Your position. You report to the Division Chief. You have five business days for the initial product. Ordinary records fees only.

9.2 The Provided Reporting

Thirty-two items, unsorted and unevaluated. Some are decisive, some are noise, at least two conflict, at least one is probably false, and at least one raises an ethical question you must handle explicitly.

#	Category	Source	Content
1	Internal	Division complaint database	212 complaints, 8 months, naming 14 distinct business names across 31 spelling variants
2	Internal	Complaint database	168 of 212 complaints name one of 6 business names
3	Internal	Complaint database	Complaint dates cluster in 3 waves, each following a different regional storm event
4	Corporate filings	State registry	6 entities matching those names; 4 active, 2 administratively dissolved in the past 5 months
5	Corporate filings	State registry, officer search	3 of the 6 list the same individual as manager or officer
6	Corporate filings	State registry	A seventh entity, not named in any complaint, shares a registered agent address and one officer

#	Category	Source	Content
			with 2 of the 6
7	Corporate filings	Registry, former names	One dissolved entity's former name matches a business named in complaints from 3 years ago in a closed file
8	Government records	State contractor licensing board	4 of the 6 hold active electrical or solar contractor licenses; 2 have never held one
9	Government records	Licensing board disciplinary records	2 prior disciplinary actions against one licensee, both resolved by consent, 4 and 2 years ago
10	Government records	County permit portals, 3 counties	Permits filed for 61% of the installation addresses appearing in complaints; 39% have no permit on file
11	Government records	County permit portals	Of permits filed, 44% show "issued, no final inspection"
12	Government records	County recorder	UCC fixture filings recorded against 340 residential parcels naming a single financing entity
13	Corporate filings	Registry, other state	The financing entity is registered in another state; officers do not overlap with the 6 installers on the record
14	Public databases	Federal and state utility interconnection records	2,100 interconnection applications from the 6 entities over 3 years; 620 show no completion status
15	Public databases	Utility interconnection records	Completion rates vary sharply: 2 entities above 90%, 3 between 60 and 75%, 1 at 31%

#	Category	Source	Content
16	Court records	State trial courts, 4 counties	9 civil cases naming the entities: 6 consumer contract claims, 2 mechanics lien foreclosures, 1 employment matter
17	Court records	Docket detail	One consumer case includes an exhibit: a door-to-door sales script filed by the plaintiff
18	Court records	Docket detail	5 of the 6 consumer cases were resolved by stipulated dismissal; terms not filed
19	Court records	Federal district court	One case: the financing entity v. an installer, breach of contract, filed 5 months ago, pending
20	Financial records	Municipal and county check registers	Two of the entities appear as vendors on small municipal solar projects in the past 3 years
21	Archived websites	Captures of 4 installer sites	All four sites listed identical warranty language, with one company's name left uncorrected in two of them
22	Archived websites	Captures of 1 installer site	A capture 14 months ago listed a "financing partner" that the current site does not name; the named partner is the financing entity from item 12
23	Internet / public databases	Domain registration data	9 domains associated with the 6 entities were registered across 3 clusters, each within a 12-day window, each preceding a complaint wave by roughly 5 weeks

#	Category	Source	Content
24	Internet / public databases	Passive DNS and hosting	6 of the 9 domains resolved at various times to the same hosting address
25	Newspapers	Local coverage, 3 outlets	Articles describing homeowner complaints, quoting named homeowners and a trade association spokesperson
26	Newspapers	Legal notices	Fictitious business name filings connecting 2 trade names to 1 of the 6 entities
27	Social media	Community groups	Numerous posts from apparent homeowners; also 5 accounts created within one month posting identical text praising one of the entities
28	Podcasts	Industry podcast, 9 months ago	An executive of one entity describes a sales model based on independent contractor crews paid on signed contracts rather than on completed installations
29	Satellite imagery	Commercial imagery over 40 complaint addresses	Panels visibly installed at 33 of 40; absent at 7, where complaints allege systems were never installed
30	Metadata	Sales script exhibit (item 17) PDF	Author field names an individual; creating application is an enterprise suite; creation date 3 years ago; document contains tracked changes removing a disclosure sentence
31	Trade association	Public statement	Asserts that the complaints stem from "a

#	Category	Source	Content
			handful of bad actors" and that licensed members are unaffected
32	Public databases	Regional commodity report	Solar module prices fell 14% over the period

9.3 Guidance on the Substance

A weak submission produces a summary of the complaints. A strong submission notices which of the three courses of action the evidence actually supports, distinguishes the entities from one another rather than treating them as a bloc, and identifies the one structural feature that explains the complaint pattern better than the others do.

Assess the trade association's claim by the same standard as the complaints. It may be partly right, and saying so plainly is the professional response.

Item 27 concerns accounts. Item 25 concerns named private individuals. Item 30 names an individual. Item 1 is a database of private individuals' complaints. Each requires an explicit handling decision, and the Division's exposure to public records law makes those decisions consequential.

9.4 What You Must Produce

One deliverable per phase. Each must document the decisions made and justify them.

Deliverable 1: Requirements Package (12 points)

- The decision statement: which of the three courses of action, who decides, by when.
- **Three PIRs**, each passing the four tests, with the failed drafts retained.
- EEIs and observable indicators for each.
- An answerability assessment naming at least three questions open sources cannot reach and what would answer each.
- A prioritization using the two-dimensional grid, with the constraint stated.

Deliverable 2: Collection Plan (12 points)

- Minimum eighteen rows, six-plus categories, five-plus from official and compelled sources.
- Every column from Section 1.3.1, with the expected coverage boundary written in advance.
- Four tests applied per row, with at least one row narrowed or removed and the reasoning recorded.

- A prohibited-activity statement and an individuals policy specific to this engagement, addressing complainants explicitly.
- Slow items identified and sequenced first.
- Stopping conditions per EEI.

Deliverable 3: Collection Record (10 points)

Using the provided packet as your collected material, plus at least six additional plan rows you would execute and what you would expect each to return:

- All 32 items logged to the Section 1.4.2 standard, with reliability and credibility ratings and justifications.
- Noise identified.
- At least four negative results you would expect, with their coverage boundaries.
- The two conflicting items identified with your resolution; the probably-false item identified with reasoning.

Deliverable 4: Processed Dataset (10 points)

- The minimum viable schema applied to all 32 items.
- Event dates separated from publication dates.
- Canonical entity list covering all 7 entities, the financing entity, and the trade names, with the basis for every merge and at least one unresolved flag.
- Duplicates marked.
- A master timeline with a precision column and supporting category counts.
- An explicit de-identification decision for complainant data, made at this phase, with the reasoning.

Deliverable 5: Exploitation Record (12 points)

- Minimum ten derived findings, across at least five operations.
- Every finding in two-column observation and interpretation form, with method and confidence.
- A cross-reference harvest with at least four references dispositioned.
- A relationship map covering entities, officers, agents, domains, and the financing entity, with the basis and supporting items for every edge, and single-source edges flagged.

Deliverable 6: Analysis Package (14 points)

- Minimum four hypotheses explaining the complaint pattern, including the trade association's, stated fairly.

- A scoring matrix with disconfirmation questions answered per hypothesis.
- Discriminating evidence identified, with the number of items the judgment rests on stated.
- **Entity-by-entity differentiation:** your assessment must distinguish among the seven entities rather than treating them as one, and confidence may differ by entity.
- Six-plus key assumptions including at least one coverage assumption.
- Confidence with category-count reasoning.
- Gaps ranked by analytic impact, each with a named resolving source.

Deliverable 7: Intelligence Product (16 points)

Maximum three pages, plus annexes. Addressed to the Division Chief, written in the knowledge that it may become a public record.

Required: bottom line (four sentences, judgment, confidence, and which course of action the evidence supports); findings by PIR with facts separated from assessments and every fact rated and categorized; entity-by-entity differentiation; assessment of the trade association's claim; assumptions; ranked gaps; at least eight recommendations that are specific, assignable, and time-bound, including at least two executable this week at no cost and at least two that are referrals or information requests rather than conclusions; what would change this assessment; collection log annex covering all 32 items; ethics annex.

Characterization discipline is graded heavily here. Complaints are allegations. Stipulated dismissals establish nothing. Two entities have never held a license, which is a records fact; whether that constitutes a violation is a legal determination you do not make.

Deliverable 8: Dissemination and Feedback Plan (8 points)

- The dissemination decisions: consumers (the Division Chief plus at least two others), format, length, channel, timing, handling, retention, and receipt confirmation for each.
- A one-paragraph flash version.
- A five-minute briefing outline following Section 1.8.4.
- A utility feedback instrument, and the accuracy record for your central forecast with a resolution date.
- Two anticipated stakeholder challenges and your written responses.

Deliverable 9: Continuous Monitoring Strategy (10 points)

- The standing requirement and the dated baseline.
- Minimum five indicators, each tested against the five properties, and each tied to something in your analysis. At least one must be a leading indicator drawn from the item-23 pattern.
- Sources, cadence, and owners.

- Three thresholds with defined actions and named owners.
- Alert fatigue, silent breakage, and baseline drift controls, each specific.
- A retention and purge schedule addressing complainant data.
- A review cycle and an explicit retirement condition.

Deliverable 10 (required, not separately scored): Decision Justification Log

A running log, one row per significant investigative decision across all nine phases: what you decided, the alternatives, why you chose as you did, and what would have changed the decision. Minimum fifteen rows. Absence of this log caps the capstone at 70 points.

9.5 Capstone Grading Rubric (110 points, scored to 100)

Deliverable	Criterion	Points
1. Requirements	Decision statement precise; three PIRs pass four tests	5
	EEIs and observable indicators sufficient	4
	Answerability assessment honest, with three-plus limits	2
	Prioritization against a stated constraint	1
2. Plan	Eighteen-plus rows, breadth and official-source requirement met	4
	Coverage boundaries written in advance	3
	Four tests per row; at least one narrowed with reasoning	3
	Prohibited-activity statement and individuals policy addressing complainants	2
3. Collection	All 32 items logged with defensible ratings	4
	Noise, conflicts, and probable falsehood identified with reasoning	3
	Six additional plan rows with expected returns	2
	Negative results with boundaries	1

Deliverable	Criterion	Points
4. Processing	Schema applied; dual dates separated	3
	Canonical entity list with merge basis and unresolved flags	4
	Master timeline with precision and category counts	2
	De-identification decision made at processing, with reasoning	1
5. Exploitation	Ten-plus findings across five-plus operations	4
	Observation versus interpretation maintained throughout	4
	Cross-reference harvest dispositioned	2
	Relationship map with edge basis; single-source edges flagged	2
6. Analysis	Four-plus hypotheses, trade association's stated fairly	4
	Disconfirmation answered per hypothesis	3
	Discriminating evidence isolated; count stated	3
	Entity-by-entity differentiation with per-entity confidence	2
	Assumptions including a coverage assumption	1
	Confidence with category-count reasoning; gaps ranked with resolving sources	1
7. Product	Bottom line: four sentences, judgment, confidence, course of action	4
	Facts separated and rated; characterization discipline correct	4
	Trade association claim assessed to the same standard	2
	Recommendations: specific,	3

Deliverable	Criterion	Points
	assignable, time-bound, including no-cost and referral options	
	Collection log and ethics annex complete	2
	Length discipline	1
8. Dissemination and feedback	Decisions complete for three-plus consumers	3
	Flash version and briefing outline	2
	Feedback instrument and falsifiable accuracy record with a resolution date	2
	Anticipated challenges answered	1
9. Continuous	Standing requirement, baseline, five-plus tested indicators including a leading one	4
	Thresholds with actions and owners	2
	Three failure-mode controls, specific	2
	Retention and purge for complainant data; review cycle and retirement condition	2
Total		110, scored to 100

Grade bands: 90 to 100 exemplary, 80 to 89 proficient, 70 to 79 competent, 60 to 69 developing, below 60 not yet passing.

Automatic deductions:

- Any judgment without a confidence level: minus 5 per instance.
- Any finding not traceable through the documentation chain: minus 3 per instance.
- Treating a complaint allegation or a stipulated dismissal as an established fact: minus 8 per instance.
- Asserting a legal violation rather than describing the record: minus 8 per instance.
- Treating the seven entities as an undifferentiated bloc in the assessment: minus 10.
- Identifying any complainant by name, address, or any combination of details that would identify

them, anywhere in any deliverable: minus 25. Complainant data is used only in aggregate and de-identified at processing, and the ethics annex must say so.

- Missing decision justification log: capped at 70.
-

Part 10: Answer Keys

Instructors may wish to remove this section from student copies.

10.1 Short Answer Key

1. Repeatability, auditability, efficiency, and diagnosability. Diagnosability is underrated because when an effort fails the instinct is to collect more, and in most documented failures the information was substantially available; named phases let an organization locate the actual break, which is usually in processing, analysis, dissemination, or the consumer relationship.
2. Requirements (PIRs, EEIs, indicators), planning (collection plan, legal review, OPSEC), collection (raw items plus log), processing (normalized indexed dataset), exploitation (derived findings), analysis (assessment with confidence, assumptions, gaps), dissemination (delivered product and confirmed receipt), feedback (revised requirements and accuracy record), continuous collection (monitoring plan and alerts).
3. Exploitation operates on a single item's content, whereas processing operates on form and analysis operates across items. Collapsed into processing, it is treated as clerical and rushed, producing confident wrong derived facts that propagate as observations. Collapsed into analysis, derived findings mix with integrative judgments, so when a reader asks how a location was established there is no documented method to point to.
4. An intelligence requirement is what the consumer needs to know, phrased as a question about the world tied to a decision ("Will X begin regional operations within 18 months?"). A collection requirement is what a collector must retrieve, phrased as a specific item from a named source ("Search the recorder's index in four counties for eight name variants over 24 months").
5. Decision, PIR, EEI, indicator, collection requirement. The decision says why anyone cares; the PIR says what would change it; the EEIs say what must be true for the PIR to resolve; the indicators say what could actually be observed; the collection requirements say where to go look.
6. Decision-linked (catches "tell me everything" requirements), answerable (catches questions about private intent), deadline-bound (catches perfect analysis delivered after the decision), consequential (catches questions whose answer changes nothing).
7. It states which parts of a requirement open sources can and cannot reach. Delivering it early lets the consumer pursue another route or change the decision framing while there is still time; delivering it at the end wastes the whole cycle and damages the relationship.
8. A grid of cost against the likelihood the source holds the answer: quick wins (cheap, likely), core effort (expensive, likely), opportunistic (cheap, unlikely), last resort (expensive, unlikely). The core

effort quadrant should absorb the bulk of the work, with quick wins done first to establish the frame and opportunistic collection filling the time while slow items are in flight.

9. Because a review after collection is a post-mortem, not a review. By then any improper collection has already occurred, the exposure exists, and the findings may be unusable. Planning is also when narrowing or removing a row costs nothing.

10. A collection line with days or weeks of latency, such as a records request, an imagery order, or an interlibrary loan. It must be initiated on day one because otherwise it arrives after the decision, and in practice slow items frequently turn out to hold the discriminating evidence.

11. Identity, geography, structured official sources, observational sources, published sources, user-generated sources, with archives throughout. Run in reverse, unverified early material anchors interpretation of everything read afterward, and the investigation becomes confidently wrong.

12. Requirement satisfied (indicators checked, EEIs resolve), saturation (new collection returns no new entity, date, or relationship), deadline reached, and blocked (remaining indicators unreachable from open sources). Not stopping conditions: feeling that you have enough, and finding something that confirms your initial theory.

13. Because its operations are formatting and consistency operations: normalizing, deduplicating, resolving entities, translating, transcribing, structuring, tagging, and indexing. The completeness test is whether a second analyst could sort by date, group by entity, and identify duplicates in your dataset without asking you anything.

14. The event date is when the described thing occurred; the publication date is when the item appeared. Collapsing them puts events in the wrong period, which corrupts sequence and therefore any causal reasoning built on the timeline.

15. Splitting one entity into several hides patterns and under-counts activity; merging several into one invents relationships. The rule is that a name match alone is never sufficient: a second identifier is required (registration number, matching address, shared officer, linking case), and the basis for every merge is recorded.

16. Observation is what was seen; interpretation is what it is judged to mean, with a confidence level. The separation lets a reviewer accept the observation while disputing the interpretation, which is the kind of disagreement a professional product must survive, and it prevents assessments from propagating as though they were evidence.

17. Every cross-reference in a document becomes a new collection plan row or is explicitly closed with a reason. It is the highest-yield operation because official documents cite other documents, and following those citations produces material that looks, to an observer, like inside access; reading documents in isolation leaves most of the available value uncollected.

18. Assemble, generate hypotheses, test by disconfirmation, identify discriminators, surface assumptions, assign confidence, rank gaps, write. Discriminator identification separates trained analysts from careful readers because most collected evidence is consistent with every hypothesis and therefore carries no weight; isolating the few items that distinguish reveals what the judgment actually rests on,

which is usually far fewer items than were collected.

19. Utility feedback arrives days after delivery and asks whether the product was read, answered the real question, and changed anything, improving format, timing, framing, and requirement quality. Accuracy feedback arrives after the forecast window closes and asks whether the assessment was right, improving calibration, source weighting, and analytic method.

20. Alert fatigue, remedied by tuning thresholds, aggregating low-tier observations, and deleting any alert type with no defined action. Silent breakage, remedied by heartbeat checks that alert on the absence of expected input. Baseline drift, remedied by a scheduled, dated baseline refresh.

10.2 Multiple Choice Key

#	Answer	Note
1	b	Locating the break is the diagnosability benefit
2	c	A question about the world, tied to a decision
3	b	A specific retrievable item from a named source
4	b	Planning; review after collection is a post-mortem
5	b	Latency-bound; must be initiated first
6	c	Identity and geography determine what can be searched
7	b	Comparability without meaning
8	b	Event date and publication date
9	c	Exploitation: extracting what is not on the item's face
10	b	Items that distinguish between hypotheses
11	b	Distinct categories and coordination cost
12	b	Did anything change
13	c	Delete it; an alert with no action is noise
14	b	Collection stops silently and is misread as inactivity
15	b	Revise the product, hold the judgment; softening is politicization

10.3 True or False Key

#	Answer	If false, why
1	False	Phases compress in depth but are never deleted; a deleted phase moves its cost downstream where it is larger
2	True	
3	False	A post-collection review is a post-mortem; the exposure and any improper collection have already occurred
4	False	A negative result with its coverage boundary is a finding with real value and must be recorded
5	False	Processing makes material comparable and adds no meaning; interpretation belongs to exploitation and analysis
6	False	A second identifier is required, such as a registration number, matching address, shared officer, or linking case
7	True	
8	False	Most evidence is consistent with several hypotheses; only a small subset discriminates
9	True	
10	True	

10.4 Critical Thinking Guidance

7.1 Fails answerable (asks about risk in the abstract and implicitly about a competitor's intent) and fails deadline-bound and consequential as written. Rewrite along the lines of: "Will [chain] open sites within our primary service area within 18 months, ahead of the September capital vote?" EEIs: regulatory and licensing filings, property and permit activity, clinical hiring. Intent is unreachable; the reframing substitutes necessary observable precursors, and what is lost is the ability to distinguish preparation that is abandoned from preparation that proceeds.

7.2 Two legitimate readings: a flood-risk PIR tied to an evacuation or pre-positioning decision, and a

water-quality or supply PIR tied to a treatment or advisory decision. The consumer must choose because only the consumer knows which decision is live; guessing produces a product answering a question nobody had.

7.3 Present the two-by-two grid with all five placed, execute the two in the high-impact and answerable quadrant, and for the unanswerable high-impact one write something like: "PIR 2 asks about the competitor's internal intent, which open sources cannot reach. We can answer whether the observable steps that would necessarily precede entry have occurred. If you need intent itself, that requires [named non-open route], and I recommend deciding that this week."

7.4 Rows should each name a category, a specific system, an exact query with name variants, a range, an access model, a legal note, and a coverage boundary written in advance. The slow item is typically a records request or an imagery order; initiated in week three it arrives after the decision, and in most investigations the slow item is disproportionately likely to hold the discriminating evidence.

7.5 The row fails the individuals policy and probably the purpose-and-harm test: it collects about identifiable individuals who are not the subject of the requirement, and the material is low-reliability conversational content. Either remove it, or narrow it to the organization's own official accounts and public statements by named officers acting in their official capacity, and say so explicitly.

7.6 Execute all quick wins and the highest-priority core rows; initiate every slow item regardless of quadrant, because initiation is cheap and latency is not; defer opportunistic rows to gap time; drop last-resort rows and record that they were dropped and what they would have addressed.

7.7 The regulatory filing is compelled, formatted, and penalized, so reliability A or B; the trade press article's sourcing is the subject's own officials, so it is not independent and credibility is 3 at best. Product sentence: "A state regulatory filing records X (A1). A trade press article attributes a differing figure to company officials (C3, not independent of the subject); we rely on the filing and note the discrepancy."

7.8 Six items in one category share a production process and its failure modes and may share an origin, so they can fail together. Three categories with unrelated production processes cannot be cheaply coordinated. Confidence wording: "Moderate confidence: supported by six items, all from a single category, with no independent corroboration from a differently produced source."

7.9 Reliability is a track record and credibility is a per-claim judgment; a reliable source can carry a low-credibility claim. Weight the compelled records, report the conflict explicitly rather than suppressing it, and record in the source's history that this claim conflicted with compelled records, without downgrading the source's overall rating on a single instance.

7.10 Normalize time zones first (prevents false event sequences and invented causation), then date formats (prevents mis-sorting), then fiscal calendars (prevents period comparison errors), then separate event dates from publication dates (prevents events landing in the wrong period). Each step is recorded in processing notes.

7.11 The first two resolve to one canonical entity on the strength of registration number or address plus name similarity. The third is a distinct entity that may be related; requires a shared officer, agent

address, or a linking filing before asserting a relationship, and it is a relationship, not a merge. The fourth is a former name and merges into whichever entity's registry record carries it. Anything without a second identifier stays flagged as unresolved and becomes a collection requirement.

7.12 Duplicates are evidence of propagation: they show how a claim spread, which outlets are derivative, and where circular reporting exists. Deleting them destroys the ability to detect circularity and makes it impossible to reconstruct why an item count looked strong.

7.13 The ratio is close to meaningless because most items are consistent with several hypotheses; support counts measure volume, not discrimination. Identify the discriminators by asking, item by item, whether it would look different under the competing hypothesis. Typically two or three items carry the judgment, and naming them is what makes the confidence level defensible.

7.14 A coverage assumption. Test it by establishing whether the jurisdiction requires such a permit for this work, whether the record is digitized for the relevant period, and whether it might be filed under another name or parcel. If untested, confidence should be capped at moderate at best, and the assumption must appear in the product with its consequence.

7.15 Confirmation bias. The control is a structured competing hypotheses pass with disconfirmation scoring, ideally after a break, plus a review of every item currently marked as supporting to check whether it is actually neutral. If the control changes the judgment, tell the consumer plainly what changed and why, including that a bias control was the cause; that disclosure builds rather than damages credibility.

7.16 A model answer contains: the judgment, the confidence with its reason, an estimative term from a published scale, and an explicit now-versus-later split, for example: "We assess with moderate confidence that [X] will occur by [date]; the judgment is supported by three independent source categories with unrelated production processes, and it is likely (55 to 80%) on the current evidence. One finding requires action this week: [Y]. The remainder informs the [date] decision and does not require action now."

7.17 Fact: "Registry filings list the same individual as manager of both entities, and both list the same principal address." Assessment: "We assess that the two entities are under common management, which is inconsistent with the representation of independence; high confidence, based on registry filings and property records." Improper: "The company concealed its ownership of the subcontractor." The third asserts intent and a wrongful act, neither of which the record establishes, and it carries defamation exposure.

7.18 A one-page brief for the six-minute consumer, containing the judgment, confidence, two or three discriminating facts, and the recommended action. A full assessment with annexes for the legal team, containing the same judgment plus complete sourcing, the characterization of every record type, the assumptions, the ranked gaps, the collection log, and the ethics annex. What is in one and not the other is the evidentiary apparatus, not the conclusion.

7.19 Four diagnoses: it was wrong; it was late; it was obvious to the consumer already; or it answered a question the consumer did not have. Question 1 (did you read it) separates the delivery failures; question 2 (did it answer the question you had) separates the requirement failure; if the answer to both

is yes and nothing changed, the finding was either already known or not decision-relevant. Remedies: analytic review, earlier delivery or interim products, requirement negotiation, and consumer engagement respectively.

7.20 Score the outcome as incorrect and score the process separately: given what was knowable, was the reasoning sound, were alternatives tested, was the confidence justified by the evidence at the time. If the reasoning was sound but a source was wrong, the system change is to adjust that source's reliability rating with the reason recorded, and to add a corroboration requirement for that source class rather than to change the analytic method.

7.21 A complete answer contains a standing requirement tied to a decision, a dated baseline, four indicators each passing observable, attributable, discriminating, timely, and sustainable, a source and cadence per indicator, three thresholds with defined actions and named owners, daily through annual review activities, and a retirement condition such as "retire when the underlying decision is made or when the concern has not recurred for four consecutive quarterly reviews."

Appendix A: Reusable Templates

A.1 Requirement Record

REQUIREMENT ID: _____ DATE: _____ ANALYST: _____
 DECISION SUPPORTED: _____
 DECISION MAKER: _____ DECISION DATE: _____
 PIR: _____
 FOUR TESTS: Decision-linked ☐ Answerable ☐
 Deadline-bound ☐ Consequential ☐
 EEIs: 1. _____ 2. _____ 3. _____ 4. _____
 INDICATORS (per EEI, observable only): _____
 ANSWERABILITY: reachable _____ | not reachable _____
 Non-open route for unreachable parts: _____
 PRIORITY: _____ of _____ REASON: _____
 ENDORSED BY: _____ DATE: _____
 REVISION HISTORY: _____

A.2 Collection Plan Row

Ind. ID	Category	Source/system	Query	Range	Access/cost	Legal note	Expected coverage boundary	Latency	Owner	Priority	Status	Output location
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A.3 Collection Log

Item ID	Ind. ID	Summary	Category	Source	URL	Query	Access date/time (TZ)	Archive	Publisher type	Independent?	Rel.	Cred.	Coverage boundary	Collector
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A.4 Processing Schema

Item ID	Ind. ID	Type	Category	Source	Event date	Pub. date	Access date/time	Entities (canonical)	Location	Duplicate of	Independent	Rel./Cred.	Archive	Processing notes
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A.5 Derived Finding Record

Finding ID	Source item ID	Operation	Method detail	OBSERVATION	INTERPRETATION	Confidence in derivation
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A.6 Competing Hypotheses Matrix

Item / finding	H1	H2	H3	H4	Discriminating?
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Per hypothesis: What would disprove it? Did I look? What did I find?

A.7 Product Skeleton

BOTTOM LINE Judgment + confidence + reason + so-what. 3-4 sentences.
 KEY FACTS Observed only. Category, source, Rel./Cred. per fact.
 ASSESSMENT Meaning. Alternatives rejected and why.
 Discriminating evidence named.
 KEY ASSUMPTIONS Each with what changes if wrong.
 INTELLIGENCE GAPS Ranked by analytic impact; resolving source named.
 RECOMMENDED ACTIONS Specific, assignable, time-bound.
 WHAT WOULD CHANGE THIS ASSESSMENT
 --- annexes (do not count toward length) ---
 COLLECTION LOG
 ETHICS ANNEX

A.8 Continuous Monitoring Plan

Element	Content
Standing requirement	
Baseline (dated)	

Element	Content
Indicator (x5)	Observable / attributable / discriminating / timely / sustainable
Source and cadence per indicator	
Tier 1 Watch: condition, action, owner	
Tier 2 Elevated: condition, action, owner	
Tier 3 Trigger: condition, action, owner	
Alert fatigue control	
Silent breakage control (heartbeat)	
Baseline drift control (refresh date)	
Retention and purge schedule	
Review cycle (daily / weekly / monthly / quarterly / annual)	
Retirement condition	

A.9 Decision Justification Log

#	Phase	Decision	Alternatives considered	Reason chosen	What would have changed it
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Appendix B: Phase Quality Gate Checklist

Print this. Work through it at each transition.

Requirements → Planning

1. Each PIR names a decision, a decision maker, and a date.
2. Each PIR is answerable from available sources, or its unanswerable part is named.
3. Each PIR would change what the consumer does.
4. Every EEI traces to a PIR; every indicator traces to an EEI and is observable.
5. The consumer has endorsed the requirement in writing.

Planning → Collection 6. Every indicator has at least one plan row. 7. Every row names a specific system and an executable query. 8. Every row has passed the four tests, with failures recorded. 9. Every row has an expected coverage boundary written in advance. 10. Slow items are identified and scheduled first. 11. The prohibited-activity statement and individuals policy exist. 12. Stopping conditions are written per EEI.

Collection → Processing 13. Every item is logged at the moment of collection, to the full standard. 14. Every item is archived, and the archive is retrievable. 15. Every negative result carries its coverage boundary. 16. Every deviation from the plan is recorded with its reason. 17. The stopping condition that applied is named per EEI.

Processing → Exploitation 18. Every item has a normalized event date and publication date, or an explicit note. 19. Every entity reference maps to a canonical entity, or is flagged unresolved. 20. Every merge has a recorded basis beyond a name match. 21. Duplicates are marked, not deleted. 22. Every item is tagged to at least one indicator. 23. The master timeline exists, with a precision column.

Exploitation → Analysis 24. Every derived finding is recorded against its source item ID. 25. Every derived finding names the method used. 26. Every derived finding carries its own confidence. 27. Observation is separated from interpretation without exception. 28. Every cross-reference is added to the plan or closed with a reason.

Analysis → Dissemination 29. At least two alternatives were written and tested, not just named. 30. Disconfirmation was attempted for each hypothesis. 31. Discriminating evidence is identified and counted. 32. Assumptions are written with consequences, including coverage assumptions. 33. Confidence has a stated reason, ideally on the category count. 34. Gaps are ranked, each with a resolving source. 35. No assessment appears as a fact.

Dissemination → Feedback 36. Format, length, and channel match the named consumer. 37. The judgment is in the first sentence. 38. Record types are characterized correctly. 39. Coverage boundaries survive into the product. 40. Receipt is confirmed.

Feedback → Continuous 41. The four utility questions have been asked and answered. 42. The forecast is logged falsifiably with a resolution date. 43. System changes arising from feedback are recorded. 44. The standing requirement, baseline, indicators, thresholds, owners, review cycle, and retirement condition exist.

Appendix C: Ethics and Self-Assessment Checklist

Authorization

1. Every source was lawfully accessible to any member of the public.
2. No identity was misrepresented.
3. No access control, paywall, or credential was defeated.
4. No material was accepted from anyone unauthorized to share it.
5. No dark web access.
6. Automation, if used, was permitted, disclosed, rate-limited, and authorized.

Individuals 7. No private individual is a subject of collection. 8. Individuals appear only in a professional or official capacity relevant to a requirement. 9. No minors, family members, or associates

were researched. 10. Personal data was de-identified or aggregated at processing, not at dissemination. 11. No aggregate profile of any person was created. 12. The ethics annex records every individual encountered, what was deliberately not collected, and the retention decision.

Evidentiary discipline 13. Every fact carries its category, source, access time, and ratings. 14. Every negative finding carries its coverage boundary. 15. Allegations, exhibits, testimony, orders, and settlements are each characterized correctly. 16. No name match is asserted as an identity match without a second identifier. 17. No legal conclusion is asserted where only a records fact exists.

Analytic discipline 18. The first sentence carries a judgment and a confidence level. 19. Facts and assessments are visibly separated. 20. Alternatives were tested by disconfirmation, and the discriminating evidence is named. 21. The same standard was applied to claims favoring and disfavoring every party. 22. Every sentence in the product survives the audit test back through the documentation chain.

Appendix D: Where to Go Next

Skills to build after this chapter, in priority order:

1. **Requirement negotiation with a real consumer.** The hardest and most valuable skill in this chapter. Practice converting a vague ask into an answerable, endorsed question with someone who actually has a decision to make.
2. **Processing at volume.** Take a dataset of several hundred mixed items and process it properly. The discipline only becomes real above the volume a person can hold in their head.
3. **Exploitation technique depth.** Geolocation, chronolocation, imagery interpretation, and document exploitation each reward deliberate drilling against known-answer material.
4. **Structured analytic techniques in practice,** especially Analysis of Competing Hypotheses run against a case where you already hold a strong prior.
5. **Briefing under questioning.** Ask someone to be genuinely adversarial for ten minutes. It is uncomfortable and it is the fastest way to find out which of your judgments you cannot defend.
6. **Calibration through a forecast log.** Begin one this week. Nothing else improves analytic quality as cheaply, and the payoff arrives in months rather than years.
7. **Monitoring program design and retirement.** Building one is common; retiring one well is rare and worth learning.

Habits to start now:

- Write the requirement down before searching, even for a personal question. Especially for a personal question.
- Log and archive at the moment of collection, always.
- For every negative result, write the coverage boundary.

- Before concluding, write the best version of the explanation you do not believe.
 - After every product, ask the consumer whether anything changed. Record the answer.
-

End of Chapter 4